Hypergraph Convolutional Network With Multiple Hyperedges Fusion for Hyperspectral Image Classification Under Limited Samples

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***Abstract*— Graph convolutional network (GCN) combined with convolutional neural network (CNN) exhibits significant potential in hyperspectral image (HSI) classification. Hypergraph convolutional network (HGCN) can address the limitations of GCN-based methods in representing high-order nonlinear relationships among multiple nodes. However, the existing pixel-based HGCN methods mostly adopt partial pixels for hypergraph modeling, thus limiting the representation of the global structure. Additionally, both pixel-based and superpixel-based HGCN methods only utilize the k-nearest neighbors (kNN) for hyperedge representation, thereby ignoring the rich topological information in HSI segmentation regions. To tackle these issues, we propose an HGCN with multiple hyperedges fusion (HGCN-MHF) for HSI classification with limited samples. First, we introduce a CNN branch for capturing spatial and spectral pixel-level features with different receptive fields, which begins with denoising and spectral transformation, followed by cross multiscale convolution (CMC). Second, we design an HGCN branch for extracting superpixel-level features guided by diversified high-order hypergraph structures, which incorporates a multiple hyperedges fusion (MHF) module followed by hypergraph convolution (HGC). Finally, a score-weighted feature fusion (SWFF) strategy is proposed to balance and promote the feature fusion of the two branches. Experimental results on four benchmark HSI datasets demonstrate that HGCN-MHF outperforms other state-of-the-art methods, with improvements in terms of overall accuracy (OA) around 3.50%–20.47% (Indian Pines), 2.85%–19.83% (University of Pavia), 2.31%–8.75% (Salinas), and 3.13%–20.03% (WHU-Hi-HongHu) under five labeled samples per class.**

***Index Terms*— Convolutional neural network (CNN), feature fusion, hypergraph convolutional network (HGCN), hyperspectral image (HSI) classification, superpixel segmentation.**

I. Introduction

HYPERSPECTRAL image (HSI) contains dozens or even hundreds of continuous narrow bands of the spectrum, ranging from visible light to short-wave infrared and even thermal infrared. These bands capture the intricate electromagnetic energy field emanating from the Earth's surface [1]. The high spectral resolution and rich spatial image information of HSIs make them suitable for determining the subtle differences between similar materials. However, their high dimension and complex spatial structure also bring new challenges to data processing [2]. HSI classification, signifying the process of assigning individual pixels to a defined set of categories [3], has recently gained popularity and has been widely applied to diverse fields, such as mineral detection [4], precision agriculture [5], and environmental monitoring [6], [7]. Compared with the rapid expansion of the availability and accessibility of hyperspectral data from satellite missions, the lack of pixel-level labeling hinders the application of fully supervised learning in HSI classification [8]. Therefore, there is an urgent need to develop more intelligent and automated methods for HSI classification to reduce manpower and material costs and improve efficiency [9].

In the past decade, deep learning (DL) techniques have developed rapidly and have been successfully introduced in HSI classification. Compared with traditional methods based on handcrafted features, DL can extract robust features from the original data through a series of hierarchical layers [10]. Specifically, convolutional neural network (CNN) exhibits powerful feature representation abilities, utilizing a set of parameters to train a kernel function for extracting specified features from the HSI. In the early works of applying CNN for HSI classification, 1-D CNN was widely used, focusing on capturing spectral features but ignoring spatial texture information [11]. Subsequently, 2-D CNN was employed to solve this issue [12]. To extract joint features from spectral and spatial space, 3-D CNN has been widely applied to HSI classification and has demonstrated better feature representation capabilities in recent years [13], [14], [15].

However, traditional CNN, with a large number of parameters, is unsuitable for HSI classification with limited samples [16]. Manual annotation is indeed laborious and time-consuming in real scenarios, which contributes to the typically very limited number of samples. To address the problem of classifying HSI with limited samples, many studies have been conducted. First, one possible solution from the perspective of model design is to develop lightweight models, which can reduce the number of model parameters

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and alleviate the problem of overfitting [17], [18]. Additionally, integrating CNN networks in models is beneficial for improving the feature extraction capability of the network under limited samples [19], [20], [21], [22]. Second, other solutions from the perspective of different learning paradigms, such as transfer learning [23], [24] and few-shot learning (FSL) [25], [26], involve leveraging previous knowledge to enhance the performance on new classification tasks with limited labeled samples. Finally, some solutions from the perspective of sample utilization, such as semisupervised learning [27], [28] and active learning [29], are considered promising approaches aimed at extracting information from a large number of unlabeled samples.

Although the CNN has shown considerable superiority in HSI classification, it still has several limitations. On the one hand, the receptive field of CNN is constrained, meaning CNNs can only extract spatial and spectral information from HSI in a short-range region. On the other hand, the CNN, which is designed for Euclidean data, is used to process a regular spatial structure, making it challenging for them to grasp the complex topological structure.

To address the aforementioned issues, the graph convolutional network (GCN) [30] has quickly attracted attention due to its widespread popularity and excellent performance. The GCN executes convolutional operations on graph structure converted into the frequency domain and has achieved remarkable results in HSI classification. GCN enables end-to-end representation learning of node feature information and structural information on non-Euclidean data, encompassing regions from short range to long range. Furthermore, GCN can efficiently utilize unlabeled samples by exploiting the graph structure of the data. Recent studies have demonstrated that GCN can excellently capture spatial and spectral information, and alleviate the problem of few training samples. In the HSI classification method based on GCN, the focus of the research is on how to convert HSI into graph structure. The graph structure consists of nodes and edges, and the methods of constructing graph nodes can be divided into pixel-based and superpixel-based methods.

1) *Pixel-Based GCN:* Using image pixels as graph nodes is the most direct and simple way to build the graph. The edges are defined by the similarity weights between nodes, forming the adjacency matrix. For example, Qin et al. [31] first encoded the HSI into a graph with adjacent pixels, and fused the spectral feature difference and spatial distance between pixels to establish an adjacency matrix for the subsequent GCN. To reduce the computation cost of GCN, Hong et al. [32] proposed a mini-batch GCN allowing to train large-scale HSI in a mini-batch fashion, and successfully explored the integration of GCN and CNN. In order to enhance the combination of spectral–spatial domain information, Yang et al. [33] proposed a GCN and transformer fusion network, which made full use of the context information of classified pixels and established the remote dependence relationship of spectral domain. Additionally, Zuo et al. [34] proposed an edge-labeling graph neural network (EGNN) for learning the edge labels from the similarity of connected nodes, marking the attempt to leverage GCN in HSI FSL.
2) *Superpixel-Based GCN:* However, when the size of an HSI is too large, constructing a global graph with all pixels as nodes may increase the time and space complexity of subsequent convolution operations. Conversely, constructing a local graph with partial pixels as nodes may limit the perception of the global structure. To address this issue, researchers proposed constructing the graph structure with superpixels as nodes. For example, Wan et al. [35] first utilized simple linear iterative clustering (SLIC) [36] to segment HSI into a series of compact superpixels, and adopted the mean spectral features of all pixels in the superpixel as the spectral feature of the corresponding superpixel. Then, the edges connected the spatial neighborhood of nodes and the weight of the edge was determined by the spectral feature similarity between superpixels to construct the graph structure. Many scholars have derived a series of improved algorithms based on this graph structure. For instance, Liu et al. [37] proposed CNN-enhanced GCN (CEGCN), where CNN and GCN branches perform feature learning on small-scale pixel-level regular regions and large-scale superpixel-level irregular regions. This fused double branches structure can solve the problem of incompatibility between CNN and GCN. Besides, many researches introduced the attention mechanism to GCN and achieved excellent results, such as weighted feature fusion of CNN and graph attention network (WFCG) [38], attention multihop graph and multiscale convolutional fusion network (AMGCFN) [39], and multiscale receptive fields graph attention neural network (MRGAT) [40]. In order to address the problem of the traditional GCN-based methods in aggregating the new nodes, Ding et al. [41] employed a multilevel graph sample and aggregate (GraphSAGE) network to capture long-range contextual relations and the information of new neighbor nodes. At the same year, Yao et al. [42] also proposed a multiscale GraphSAGE to learn the global and contextual information, and effectively correct errors in input graph structure. To solve the difficult problem of HSI classification in limited labeled samples, our previous work [43] proposed a differentiated-scale restricted GCN (DSR-GCN) and improved a lightweight spectral–spatial Siamese network (S3Net) [44], fused with GCN results of different size levels to capture more discriminative features in FSL scenarios. Additionally, Ding et al. [45] introduced a deep hybrid multigraph neural network (DHMG), incorporating three different graph filters to alleviate the problem of oversmoothing existing in traditional GCNs, and applying to limited number of labeled samples.

While the aforementioned studies have demonstrated the effectiveness of GCN for HSI classification, there are still notable limitations in GCN methods. For example, GCNs

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utilize low-order pairwise connections among nodes in the graph structure. However, ground objects usually possess high-order relationships that surpass the capabilities of graph structures to formulate such connections.

To tackle this challenging issue, the hypergraph neural network (HGNN) [46] is proposed, employing the hypergraph structure for data modeling. The main distinction between hypergraph and graph lies in the definition of edges. In the hypergraph structure, hyperedges are used to define the connection relationships between nodes, and each hyperedge can connect more than two nodes. In contrast, edges in a graph can only represent pairwise relationships, i.e., node-to-node binary correlations. At present, there are few studies on HGNN for HSI classification. To describe the potential complex nonlinear high-order relationships in HSI data, Ma et al. [47] proposed a feature fusion hypergraph neural network (F^2HNN), which represents each pixel as a vertex and utilizes hyperedges generated from the k-nearest neighbors (kNN) for hypergraph convolution (HGC) in the frequency domain. Duan et al. [48] also adopted this kind of hypergraph construction strategy and built a structure preservation layer to optimize the distribution of convolutional features under the guidance of hypergraph. To effectively and efficiently extract the complex spectral–spatial structure feature of HSI, Xu et al. [49] designed a concise hypergraph convolutional network (HGCN), using segmented superpixels as nodes and constructing hyperedges from kNN. In the same year, Xu et al. [50] also proposed dynamically updating the hypergraph model based on this construction strategy to optimize the hypergraph structure using the knowledge learned by the network.

Nevertheless, current HGNN-based methods still have some shortcomings. On the one hand, the existing pixel-based HGNN methods only adopt labeled pixels for constructing the hypergraph, limiting the representation of the global structure. On the other hand, both pixel-based and superpixel-based HGNN methods only utilize the kNN of the central node in the spectral feature space for hyperedge representation, overlooking the rich topological information in the HSI segmented region.

To address these issues, we propose an HGCN with multiple hyperedges fusion (HGCN-MHF) for HSI classification under limited samples. Internally, we initially introduce a CNN branch incorporating denoising and spectral transformation followed by cross multiscale convolution (CMC). This branch employs 1-D convolutions for spectral feature transformation and 2-D convolutions with different size kernels for spatial texture information acquisition to capture pixel-level features. Subsequently, we design an HGCN branch, which consists of a multiple hyperedges fusion (MHF) module and HGC. This branch is engineered to capitalize on the spatial topological information and spectral features inherent within segmented regions to create a comprehensive hypergraph and extract superpixel-level features. Finally, we propose a score-weighted feature fusion (SWFF) strategy, combining the features obtained from the two branches and their concatenated feature using adaptive weighting based on calculated scores, and then the softmax function is applied to predict classification map.

Overall, the main contributions of this article are outlined as follows.

1) In order to better represent the nonlinear high-order relationships in HSI, we propose an MHF method considering both spectral information and topological structure in segment regions to generate diverse hyperedges from different perspectives.
2) We design a multiscale CNN branch in a cross path to capture fixed size kernels in pixel-level features. Furthermore, we introduce an HGCN branch based on spatial domain HGC to extract superpixel-level features.
3) For the purpose of balancing and promoting the feature fusion of two branches, an SWFF strategy is designed to adaptively fuse features and improve the accuracy of classification results.

The main differences between the proposed HGCN-MHF method and other HGNN-based methods [47], [48], [49], [50] are given as follows.

1) The hyperedge construction in the existing HGNN methods is restricted by only employing the kNN strategy. On the contrary, we innovatively construct a hypergraph with MHF, which considers not only the spectral similarity between nodes, but also the spatial topological relationship of the segmentation region.
2) Existing HGNN-based methods all use HGC-based frequency domain. On the contrary, we explore the use of HGC based on spatial domain.

The rest of this article is organized as follows. The proposed HGCN-MHF is introduced in Section II. The experiments are illustrated in Section III. The discussions are exhibited in Section IV. The conclusion is presented in Section V.

II. Proposed Methodology

In this section, we provide a detailed introduction to the proposed HGCN-MHF. The complete architecture of our model is illustrated in Fig. 1. The network comprises two branches, one for extracting pixel-level features, namely CNN branch, and the other for extracting superpixel-level features, namely HGCN branch. Finally, the features from the two branches are fused. In the following, we will provide detailed descriptions of each module.

A. Denoising and Spectral Transform (DST)

Original HSI data contains abundant spectral information, serving as a double-edged sword. On the one hand, abundant information enhances the capacity to distinguish between classes compared to multispectral image. On the other hand, it may also contain noise, leading to classification errors and imposing high computational costs. To address this issue, we employ two 1×1 convolutional layers for denoising and dimension reduction, aiming to reduce the parameters for enhancing robustness and stability of the model. Specifically, by inputting the feature $X^{l-1}_{i,j,k}$ into the spectral convolutional layer, we have

$$X^{l}_{i,j,m} = \sigma\left(\sum_{k} W^{l}_{i,j,m,k}\mathrm{BN}\left(X^{l-1}_{i,j,k}\right) + B^{l}_{i,j,m,k}\right) \quad (1)$$

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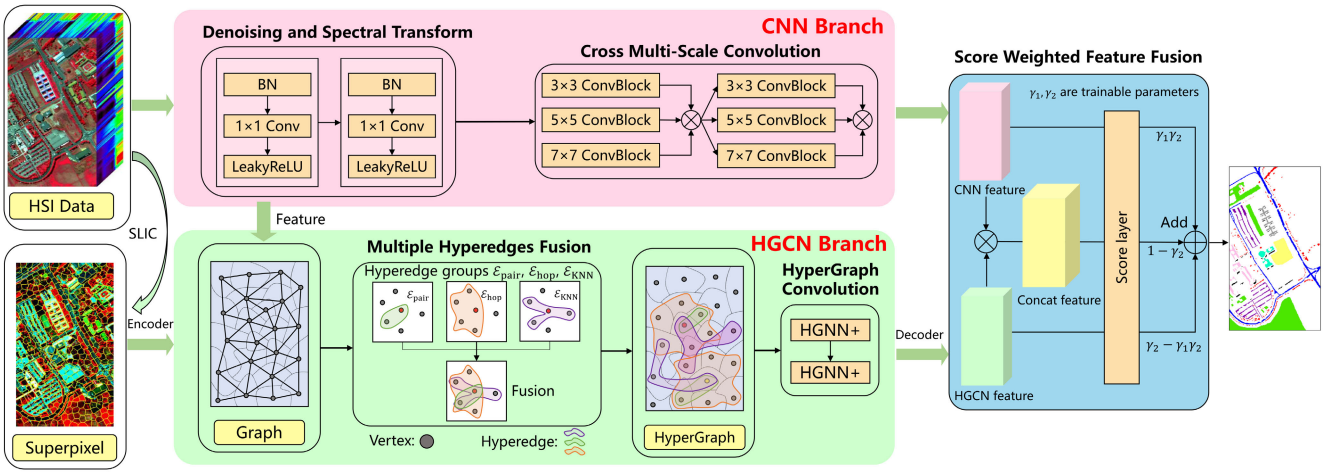


Fig. 1. Architecture of the proposed HGCN-MHF.

where $X^{l-1}_{i,j,k}$ denotes the input feature of the lth layer with spatial location (i, j) and channel position k. $X^{l}_{i,j,m}$ denotes the output feature map of the lth layer with spatial location (i, j) and channel position m. BN$(\cdot)$ denotes a batch normalized layer for reducing parameter initialization requirements and speeding up network training. $W^{l}_{i,j,m,k}$ and $B^{l}_{i,j,m,k}$ represent the weights in the convolution kernel with a size of 1×1 and the bias of the convolution, respectively. $\sigma(\cdot)$ denotes the activation function, such as LeakyReLU$(\cdot)$.

B. Cross Multiscale Convolution

To enhance the feature capture capability of CNN-based networks, a CMC is designed with kernel sizes of 7×7, 5×5, and 3×3 to generate pixel-level features. CMC consists of two independent multiscale convolutional layers, each comprising three ConvBlocks with different receptive fields. After each multiscale convolutional layer, a concatenation operation is applied to combine features with different kernel sizes, facilitating cross feature fusion.

In a $k \times k$ ConvBlock, batch normalization (BN) first adjusts the data distribution at each layer to accelerate the training speed and inhibit overfitting. To prevent the concatenated feature from exhibiting high spectral dimensions, a 1×1 convolutional layer, as specified in (1), is employed to transform the feature dimensions and reduce the number of model parameters after BN layer. Subsequently, the LeakyReLU$(\cdot)$ activation function is applied. To reduce information redundancy, mitigate the difficulty of network optimization, and prevent network overfitting, an average pooling layer is inserted. The average pooling layer uses a pooling window of size 3×3, a stride of 1 for each movement, and a padding size of 1 to maintain the image size. After average pooling, a 2-D CNN with a $k \times k$ kernel size is utilized to extract local spatial features

$$X^{l}_{i,j} = \sigma\left(\sum_{h=0}^{k-1}\sum_{w=0}^{k-1} W^{l}_{h,w} X^{l-1}_{i+h,j+w} + B^{l}_{i,j}\right) \tag{2}$$

where $X^{l}_{i,j}$ denotes the output feature of the lth layer at spatial position (i, j). $W^{l}_{h,w}$ represents the weight of the kernel and $B^{l}_{i,j}$ is the bias. $X^{l-1}_{i+h,j+w}$ is the input feature of the lth layer after average pooling operation with spatial position $(i+h, j+w)$. The final activation function $\sigma(\cdot)$ is the LeakyReLU$(\cdot)$ applied to the $k \times k$ ConvBlock.

C. Multiple Hyperedges Fusion

The HGCN branch for extracting superpixel-level features comprises two procedures: hypergraph construction and HGC. Traditional hypergraph learning approaches in HSI classification heavily rely on the kNN strategy for generating hyperedges, emphasizing spectral features but often overlooking the spatial richness inherent in the topological structure of segmented region. Additionally, due to the presence of noise in the HSI, and mixed pixels in the segmentation region, the spectral features of superpixels based on segmentation are distorted. Therefore, the misrepresentation of the relationship between ground objects is caused by relying solely on spectral features to generate hyperedges. In addition, the construction of hyperedges plays a pivotal role in determining the hypergraph structure, influencing the network's feature extraction capabilities, and consequently, the overall model performance.

To overcome this challenge, we design MHF module for hypergraph construction. In this module, three groups of strategies for generating hyperedges are adopted, namely pairwise hyperedges ($\varepsilon_{\text{pair}}$), k-hop hyperedges (ε_{hop}), and kNN hyperedges (ε_{kNN}). The strategies of $\varepsilon_{\text{pair}}$ and ε_{hop} are applicable to data with graph structure, containing both nodes and connecting edges. But the ε_{kNN} only requires to have superpixel nodes. After generating all hyperedges, we design a weight function to fuse them, considering both feature similarity and spatial distance.

First, the SLIC algorithm segments HSI data into superpixels, treating each superpixel as a node. Specifically, an association matrix $Q \in \mathbb{R}^{HW \times Z}$ determines a mapping association between pixels and superpixels, where Z denotes the number of superpixels, and HW is the number of pixels in the HSI data $X \in \mathbb{R}^{H \times W \times C}$. Then we have

$$Q_{i,j} = \begin{cases} 1, & \text{if } \hat{X}_i \in S_j \\ 0, & \text{otherwise} \end{cases} \quad \hat{X} = \text{Flatten}(X) \tag{3}$$

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where Flatten$(\cdot)$ denotes flattening X spatially. $\hat{X}_i$ represents the ith pixel in $\hat{X}$, and S_j denotes the jth superpixel. Then, we can use an encoder and decoder to achieve feature conversion, which are defined as follows:

$$V = \text{Encoder}(X; Q) = \hat{Q}^T \hat{X} \tag{4}$$

$$X_{\text{new}} = \text{Decoder}(V; Q) = \text{Reshape}(QV) \tag{5}$$

where $\hat{Q}^T$ denotes the transpose of Q after the normalized columns. V denotes the feature matrix of all superpixel nodes. Reshape$(\cdot)$ restores the spatial dimension of $\hat{X}$. X_{new} is the transformed pixel-level features derived from superpixel-level features.

Subsequently, the superpixels lack undirected edges after segmentation, rendering them nongraph structures. Therefore, undirected edges are introduced to connect adjacent superpixels in spatial distribution, transforming the data with nodes and edges into a graph structure represented as $G = (V, \epsilon)$, where G, V, and ϵ denote a graph, nodes, and edges, respectively. After constructing the graph, MHF utilizes three strategies to generate hyperedges. The first strategy is low-order correlated $\varepsilon_{\text{pair}}$, which directly converts all low-order edges in the graph into a set of two-uniform hyperedges. Each hyperedge contains only two nodes, which is the same method of edge generation in graph learning, connecting the central superpixel with its spatial neighbors, providing the necessary basic information for the higher order relationship. The second strategy is ε_{hop}, which aims to utilize a hyperedge to connect the central node to all of its kth spatial neighbors, meaning that 1-hop refers to all the neighbors of the central node, 2-hop refers to all the neighbors of the neighbor nodes of the central node, and so on. ε_{hop} can provide high-order correlation information compared to $\varepsilon_{\text{pair}}$. The third strategy is ε_{kNN}, which targets on connecting the central node and its KNN nodes in spectral feature space with a hyperedge. It is essential to note that our ε_{kNN} differs from the typical kNN used in other hypergraph generation methods. In our approach, kNN selects neighboring nodes within a specific spatial range, such as 2-hop, rather than the global range. The benefits of this method will be discussed in Section IV. The superpixel nodes in the graph are selected as the central nodes in order, and three hyperedge generation strategies are employed for each central node, resulting in three groups of hyperedges. Then, the weight of each hyperedge is calculated by measuring the similarity between the connected nodes as the fusion coefficient, and is integrated into a group of hyperedges. A hypergraph can be defined as $\mathbb{G} = (V, \varepsilon, W)$, where $\mathbb{G}$, V, ε, and W denote a hypergraph, nodes, hyperedges, and the diagonal weight matrix of hyperedges, respectively.

After all hyperedges are generated, they are fused into one group by calculating the weight of all hyperedges. Specifically, the radial basis function (RBF) measures the average similarity of the nodes connected by a hyperedge to determine the weight of this hyperedge. Due to the more similar spectra of each node, there is a greater probability that they belong to the same class. Therefore, the greater the mean spectral similarity of the nodes connected by hyperedge, the greater its weight. Additionally, spatial constraints should be introduced [51], as the greater the distance between nodes in spatial space, the lower the probability that they belong to the same class. The spectral similarity Φ_{spectral} between the nodes V_i and V_j is defined as follows:

$$\Phi_{\text{spectral}}(V_i, V_j) = \exp\left(-\|F_i - F_j\|_2^2 / (2\sigma_1^2)\right) \tag{6}$$

where $\|\cdot\|_2$ denotes the L2 norm. F_i and F_j denote the spectral features of nodes V_i and V_j, respectively. σ_1 is the adjustable parameters in the feature similarity measurement function. Similarly, the spatial similarity Φ_{spatial} is defined as follows:

$$\Phi_{\text{spatial}}(V_i, V_j) = \exp\left(-\|P_i - P_j\|_2^2 / (2\sigma_2^2)\right) \tag{7}$$

where P_i and P_j denote the central positions of superpixel nodes V_i and V_j, respectively. σ_2 is the adjustable parameters in the spatial distance measurement function. A hyperedge may connect multiple nodes. The spectral measurement weight and spatial measurement weight of hyperedge ε_k is designed as follows:

$$W_{\text{spectral}}(\varepsilon_k) = \frac{\sum \Phi_{\text{spectral}}(V_i, V_j)}{C(|\varepsilon_k|, 2)} \tag{8}$$

$$W_{\text{spatial}}(\varepsilon_k) = \frac{\sum \Phi_{\text{spatial}}(V_i, V_j)}{C(|\varepsilon_k|, 2)} \quad V_i, V_j \in \varepsilon_k,\ i \neq j \tag{9}$$

where $|\varepsilon_k|$ represents the number of nodes in the hyperedge, and $C(|\varepsilon_k|, 2)$ indicates the number of combinations of two elements chosen from $|\varepsilon_k|$ distinct elements. Finally, the weight of hyperedge ε_k is calculated as follows:

$$W_{\text{spe-spa}}(\varepsilon_k) = \text{Min}\left(W_{\text{spectral}}(\varepsilon_k), W_{\text{spatial}}(\varepsilon_k)\right) \tag{10}$$

where Min$(\cdot)$ selects the smaller value of the two as the weight, meaning only when the space is close and the spectrum is similar, the weight will be high. It is notable that RBF sets the similarity measurement between [0, 1], contributing to controlling the weight of hyperedges, ranging from 0 to 1.

D. Hypergraph Convolution

HGNNConv [46] is rooted in the spectral domain, and while it has a solid theoretical foundation, it exhibits several limitations. These shortcomings include the inability to effectively handle directed graphs, poor scalability, and the incapability to process large graphs. However, methods based on the spatial domain provide an intuitive definition, lower computational requirements, flexibility, and improved scalability. HGNNConv$^+$ [52] represents a spatial convolution method applied to hypergraphs, utilized in this study.

Graph convolution in the spatial domain aggregates features from neighboring vertices, updating the central vertex feature in the single stage. Different from graph convolution, HGNNConv$^+$ conducts a two-stage transformation, involving vertex-hyperedge–vertex interactions. As depicted in Fig. 2, HGNNConv$^+$ consists of two stages: first, hyperedge features are generated based on their associated vertices; second, vertex features are updated by all the hyperedges to which they belong. In a hypergraph $\mathbb{G} = (V, \varepsilon, W)$, $|V|$ is the number of vertices and $|\varepsilon|$ is the number of hyperedges. The incidence

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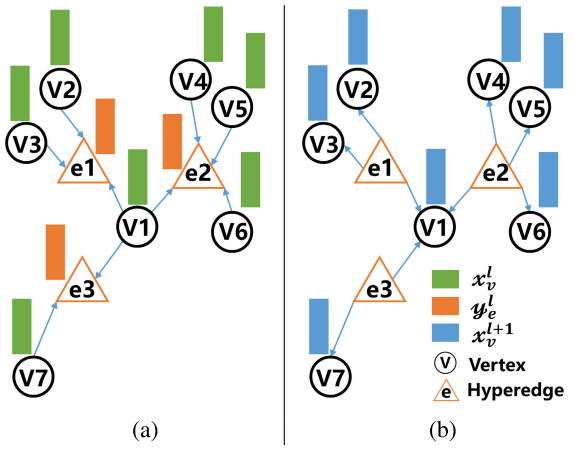


Fig. 2. Illustration for HGNNConv$^+$ [52] based on spatial domain. (a) In stage 1, the original node features x_v^l are aggregated into hyperedge features y_e^l. (b) In stage 2, hyperedge features are redistributed to the contained nodes for generating new node features x_v^{l+1}.

matrix $H \in \mathbb{R}^{|V|\times|\varepsilon|}$ represents interneighbor relation in a hypergraph and is defined as follows:

$$H(v,e) = \begin{cases} 1, & \text{if} \quad v \in e \\ 0, & \text{if} \quad v \notin e \end{cases} \tag{11}$$

where $v \in e$ denotes the vertex v connected by hyperedge e. The vertex interneighbor set $N_v(e)$ refers to all nodes connected by the hyperedge e. Analogously, the hyperedge interneighbor set $N_e(v)$ refers to all the hyperedges containing the node v. Then, a general spatial HGC can be defined as follows:

$$\begin{cases} m_e^l = \sum\limits_{v\in N_v(e)} M_v^l\left(x_v^l\right) \\ y_e^l = U_e^l\left(w_e, m_e^l\right) \end{cases} \quad \text{stage 1} \tag{12}$$

$$\begin{cases} m_v^{l+1} = \sum\limits_{e\in N_e(v)} M_e^l\left(x_v^l, y_e^l\right) \\ x_v^{l+1} = U_v^l\left(x_v^l, m_v^{l+1}\right) \end{cases} \quad \text{stage 2} \tag{13}$$

where x_v^l denotes the feature of node v in the lth layer, and m_e^l is the message of hyperedge e. w_e denotes the weight of hyperedge e, and y_e^l is the feature of hyperedge e. m_v^{l+1} is the message of vertex v, and x_v^{l+1} is the updated ouput feature of vertex v. M_v^l, U_e^l, M_e^l, and U_v^l denote the vertex message function, hyperedge update function, hyperedge message function, and vertex update function, respectively, which can be defined flexibly for specific applications. HGNNConv$^+$ specifically defines the message–update functions M_v^l, U_e^l, M_e^l, and U_v^l as follows:

$$\begin{cases} M_v^l\left(x_v^l\right) = \dfrac{x_v^l}{|N_v(e)|} \\ U_e^l\left(w_e, m_e^l\right) = w_e \times m_e^l \\ M_e^l\left(x_v^l, y_e^l\right) = \dfrac{y_e^l}{|N_e(v)|} \\ U_v^l\left(x_v^l, m_v^{l+1}\right) = \sigma\left(m_v^{l+1}\cdot\Theta^l\right) \end{cases} \tag{14}$$

where $|N_v(e)|$ denotes the number of vertexes connected by the hyperedge e, and $|N_e(v)|$ denotes the number of hyperedges containing the vertex v. The message functions $M_v^l(x_v^l)$ and $M_e^l(x_v^l, y_e^l)$ represent the role of normalization. $\sigma(\cdot)$ is an activation function like LeakyReLU$(\cdot)$. Θ^l is a trainable parameter matrix of layer l.

HGC, as formulated in (12)–(14), generates the feature of hyperedge and vertex in sequence. However, our data, composed of a hypergraph structure, lacks natural ordering, and this sequential information update pattern leads to relatively slow forward propagation. Therefore, the matrix format of HGC is proposed to address these problems. In (11), H defines hypergraph interneighbor relation. It is utilized to calculate vertex degree matrix D_v and hyperedge degree matrix D_e, which can be calculated as follows:

$$\begin{cases} D_v = \text{diag}\left(\sum\limits_{j=1}^{|\varepsilon|} H(i,j)\right), & i = 1,2,\ldots,|V| \\ D_e = \text{diag}\left(\sum\limits_{i=1}^{|V|} H(i,j)\right), & j = 1,2,\ldots,|\varepsilon| \end{cases} \tag{15}$$

where diag$(\cdot)$ denotes the convert vector to diagonal matrix. Thus, in stage 1, the hyperedge feature Y^l in the lth layer can be formulated as follows:

$$Y^l = W D_e^{-1} H^T V^l \tag{16}$$

where V^l denotes the superpixel vertex feature as the input of the lth convolution layer, and W denotes the weight matrix of hyperedges $W_{\text{spe}-\text{spa}}$. Similarly, in stage 2, the updated vertex feature V^{l+1} in the lth layer as the output can be formulated as follows:

$$V^{l+1} = \sigma\left(D_v^{-1} H Y^l \Theta^l\right). \tag{17}$$

Thus, the matrix format of HGNNConv$^+$ can be combined into one formula

$$V^{l+1} = \sigma\left(D_v^{-1} H W D_e^{-1} H^T V^l \Theta^l\right). \tag{18}$$

In contrast to HGNNConv, frequency domain-based HGNNConv$^+$ is defined as follows:

$$V^{l+1} = \sigma\left(D_v^{-\frac{1}{2}} H W D_e^{-1} H^T D_v^{-\frac{1}{2}} V^l \Theta^l\right). \tag{19}$$

The HGC consists of two layers of HGNNConv$^+$.

When a hyperedge connects only two nodes ($\varepsilon_{\text{pair}}$), HGNNConv$^+$ can be mathematically proven to be a special case of GCN. In this scenario, the simple hypergraph can be represented by a graph with an adjacency matrix A and vertex degree matrix D. In GCN, where each hyperedge is considered to have the same weight, we can express it as follows:

$$\begin{cases} D_v = D \\ HH^T = A + D \\ D_e^{-1} = \dfrac{1}{2} I \\ W = I. \end{cases} \tag{20}$$

Consequently, the HGNNConv$^+$ can be simplified as follows:

$$\begin{aligned} V^{l+1} &= \sigma\left(D_v^{-1} H W D_e^{-1} H^T V^l \Theta^l\right) \\ &= \sigma\left(D^{-1} H\left(\frac{1}{2} I\right) H^T V^l \Theta^l\right) \end{aligned}$$

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$$= \sigma\left(\frac{1}{2}D^{-1}\left(HH^T\right)V^l\Theta^l\right)$$
$$= \sigma\left(D^{-1}(A+D)V^l\hat{\Theta}^l\right)$$
$$= \sigma\left(D^{-\frac{1}{2}}\hat{A}D^{-\frac{1}{2}}V^l\hat{\Theta}^l\right) \quad (21)$$

where $\hat{A} = A + D$ and $\hat{\Theta}^l = (1/2)\Theta^l$. The extra $(1/2)$ term can be absorbed by the learnable parameter Θ. Therefore, HGNNConv$^+$ inherits the powerful expressive ability of GCN in handling simple graphs and also possesses the capability to model and learn high-order correlations in hypergraphs.

E. Score-Weighted Feature Fusion

After obtaining pixel-level CNN features and superpixel-level HGCN features, which are separately extracted by CNN branch and HGCN branch, an SWFF method is proposed to fuse these features. The score-weighted [53] strategy empowers model with a stronger data representation ability than feature-weighted addition and feature concatenation. In this section, we use F_{CNN} and F_{HGCN} to represent feature vectors extracted by CNN branch and HGCN branch, respectively. A linear layer, acting as the score layer, is employed to adjust the channel length of the feature vectors to match the number of categories. The score vector is represented by S, where S_{CNN} and S_{HGCN} are generated from F_{CNN} and F_{HGCN}, respectively, through the score layer. The value of the neurons in the score vector can determine the likelihood that the pixel will be divided into the corresponding ground class.

1) Feature Concatenation: The concatenated feature F_{concat} is acquired by concatenating F_{CNN} and F_{HGCN} in channel dimension, and it can be defined as follows:

$$F_{\text{concat}} = \text{Concat}(F_{\text{CNN}}, F_{\text{HGCN}}) = \begin{bmatrix} F_{\text{CNN}} \\ F_{\text{HGCN}} \end{bmatrix}. \quad (22)$$

S_{concat} is the score vector of F_{concat} and can be calculated as follows:

$$S_{\text{concat}} = WF_{\text{concat}} + B$$
$$= \begin{bmatrix} W_1 & W_2 \end{bmatrix}\begin{bmatrix} F_{\text{CNN}} \\ F_{\text{HGCN}} \end{bmatrix} + B_1 + B_2$$
$$= W_1F_{\text{CNN}} + B_1 + W_2F_{\text{HGCN}} + B_2 \quad (23)$$

where W and B are weight and bias of the linear layer, respectively. The matrix W consists of W_1 and W_2, which are the weights of F_{CNN} and F_{HGCN}, respectively. The matrix B consists of B_1 and B_2, which are the biases of F_{CNN} and F_{HGCN}, respectively.

2) Feature-Weighted Addition: The combined feature F_{add} obtained through feature-weighted addition incorporates a learnable weight parameter γ, which is controlled by the sigmoid function and ranges between (0,1). Similar to feature concatenation, F_{add} and S_{add} are defined as follows:

$$F_{\text{add}} = \gamma F_{\text{CNN}} + (1-\gamma)F_{\text{HGCN}} \quad (24)$$
$$S_{\text{add}} = WF_{\text{add}} + B$$
$$= \gamma WF_{\text{CNN}} + (1-\gamma)WF_{\text{HGCN}} + B. \quad (25)$$

3) Feature Score Weighted: The score vector S_{score_1} adaptively adds the score vectors of the two branches S_{CNN} and S_{HGCN} in a weighted way, and is given as follows:

$$S_{\text{score}_1} = \gamma S_{\text{CNN}} + (1-\gamma)S_{\text{HGCN}}$$
$$= \gamma(W_1F_{\text{CNN}} + B_1)$$
$$+ (1-\gamma)(W_2F_{\text{HGCN}} + B_2)$$
$$= \gamma W_1F_{\text{CNN}} + (1-\gamma)W_2F_{\text{HGCN}}$$
$$+ \gamma B_1 + (1-\gamma)B_2. \quad (26)$$

Equation (23) illustrates that S_{concat} involves two learnable matrices, W_1 and W_2, potentially offering greater expressiveness compared to feature-weighted addition. Equation (25) demonstrates S_{add}, while incorporating a learnable weight parameter γ to emphasize the differing importance of the two branches, utilizes only one learnable matrix. However, according to (26), S_{score_1} not only preserves two distinct learnable matrices but also incorporates the learnable weight parameter. Consequently, the fusion feature generated by feature score weighting exhibits a stronger data representation capability compared to those generated by feature concatenation and feature-weighted addition.

4) Score-Weighted Feature Fusion: The proposed score vector S_{score_2} adaptively weighted adds the score vectors of S_{score_1} and S_{concat}, and contains two trainable parameters γ_1 and γ_2. The formulation of S_{score_2} is defined as follows:

$$S_{\text{score}_2} = \gamma_2 S_{\text{score}_1} + (1-\gamma_2)S_{\text{concat}}$$
$$= \gamma_2(\gamma_1 S_{\text{CNN}} + (1-\gamma_1)S_{\text{HGCN}})$$
$$+ (1-\gamma_2)S_{\text{concat}}$$
$$= \gamma_1\gamma_2 S_{\text{CNN}} + (\gamma_2 - \gamma_1\gamma_2)S_{\text{HGCN}}$$
$$+ (1-\gamma_2)S_{\text{concat}}. \quad (27)$$

As illustrated in Fig. 1, SWFF combines the feature score-weighted values of the CNN and HGCN features, and includes the score of the concatenated feature in the score-weighted addition. The CNN feature, HGCN feature, and concatenated feature are each transformed into importance scores for the corresponding classes after passing through the score layer. The score layer comprises a linear layer with an output dimension equal to the number of classes, followed by a softmax$(\cdot)$ function to ensure that the sum of the importance values is 1. Subsequently, the score vectors for these features are weighted and summed. We design two trainable parameters γ_1 and γ_2, with values ranging from 0 to 1, to ensure that the sum of the weights of the three scores equals 1. The sum of scores is directly used for classification by applying the argmax$(\cdot)$ function. Equation (27) demonstrates the proposed SWFF possessing a stronger feature representation than S_{score_1}.

III. Experimental Results

In this section, four popular HSI datasets were adopted to verify the proposed method: they are Indian Pines, University of Pavia, Salinas, and WHU-Hi-HongHu.

A. Hyperspectral Datasets

1) Indian Pines: The first dataset was gathered by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Indian Pines test site in North-western Indiana. The

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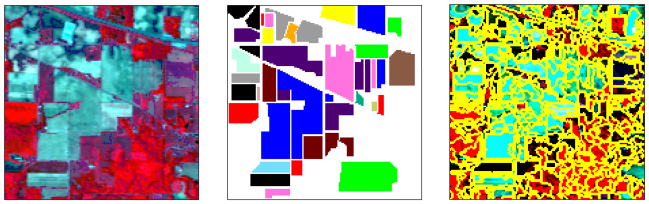


Fig. 3. Indian Pines. (a) False-color composite image. (b) Ground-truth map. (c) Segmentation result.

TABLE I
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR INDIAN PINES

No	Color	Name	Train.	Val.	Test.
1		Alfalfa	5	1	40
2		Corn-notill	5	1	1422
3		Corn-mintill	5	1	824
4		Corn	5	1	231
5		Grass-pasture	5	1	477
6		Grass-trees	5	1	724
7		Grass-pasture-moved	5	1	22
8		Hay-windrowed	5	1	472
9		Oats	5	1	14
10		Soybeans-notill	5	1	966
11		Soybeans-mintill	5	1	2449
12		Soybeans-clean	5	1	587
13		Wheat	5	1	199
14		Woods	5	1	1259
15		Bldg-grass-tree-drivers	5	1	380
16		Stone-streel-towers	5	1	87
		Total	80	16	10153

dataset consists of 145 × 145 pixels with a moderate spatial resolution of 20 m. After removing 20 water absorption bands, the dataset contains 200 remaining bands in the spectral range from 0.4 to 2.5 μm. It comprises 10 366 samples distributed among 16 classes for research and analysis. The false-color composite of HSI and the corresponding ground-truth map are shown in Fig. 3. The training and test samples are reported in Table I.

2) University of Pavia: The second dataset was collected by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor over the University of Pavia in northern Italy. The dataset has a size of 610 × 340 pixels with a high spatial resolution of 1.3 m. It comprises 103 wavelength bands in the spectral range from 0.43 to 0.86 μm, with a total of 42 776 samples distributed among nine classes for research and analysis. The false-color composite of HSI and the corresponding ground-truth map are shown in Fig. 4. The training and test samples are reported in Table II.

3) Salinas: The third dataset utilized in this study was collected by the AVIRIS sensor over the Salinas Valley in California. The dataset has a size of 512 × 217 pixels with a high spatial resolution of 3.7 m. It consists of 204 bands in the spectral dimension, covering the wavelength range from 0.4 to 2.5 μm. The dataset includes a total of 54 129 samples distributed among 16 classes for research and analysis. The false-color composite of HSI and the corresponding ground-truth map are shown in Fig. 5. The training and test samples are reported in Table III.

4) WHU-Hi-HongHu: The fourth dataset was gathered by the Headwall Nano-Hyperspec sensor on an unmanned aerial vehicle (UAV) platform over Honghu City in Hubei province, China. The dataset consists of 940 × 475 pixels, boasting a very high spatial resolution of 0.043 m. It encompasses 270 bands in the spectral dimension, covering the wavelength range from 0.4 to 1.0 μm. The dataset is composed of 386 693 samples distributed among 22 classes for research and analysis. The false-color composite of HSI and the corresponding ground-truth map are shown in Fig. 6. The training and test samples are reported in Table IV.

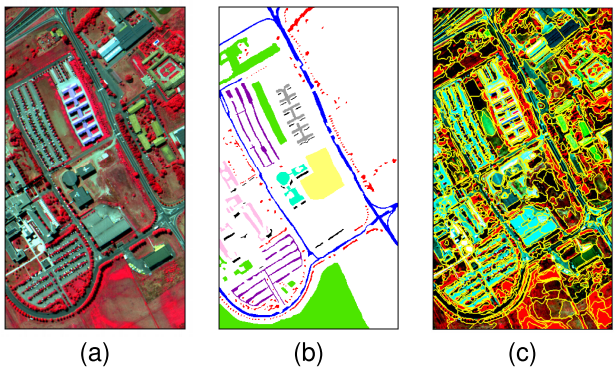


Fig. 4. University of Pavia. (a) False-color composite image. (b) Ground-truth map. (c) Segmentation result.

TABLE II
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR UNIVERSITY OF PAVIA

No	Color	Name	Train.	Val.	Test.
1		Asphalt	5	1	6625
2		Meadows	5	1	18643
3		Gravel	5	1	2093
4		Trees	5	1	3058
5		Painted metal sheets	5	1	1339
6		Bare soil	5	1	5023
7		Bitumen	5	1	1324
8		Self-Blocking Bricks	5	1	3676
9		Shadows	5	1	941
		Total	45	9	42722

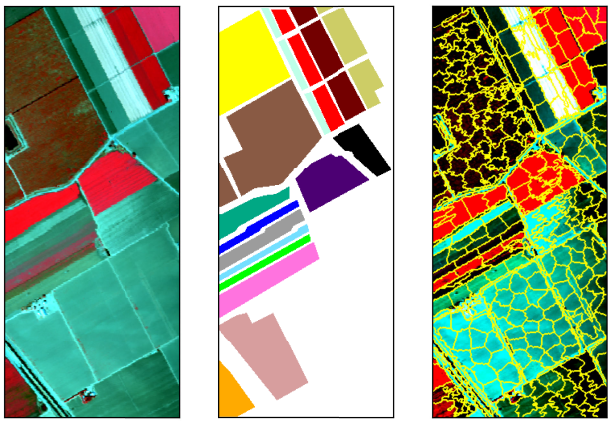


Fig. 5. Salinas. (a) False-color composite image. (b) Ground-truth map. (c) Segmentation result.

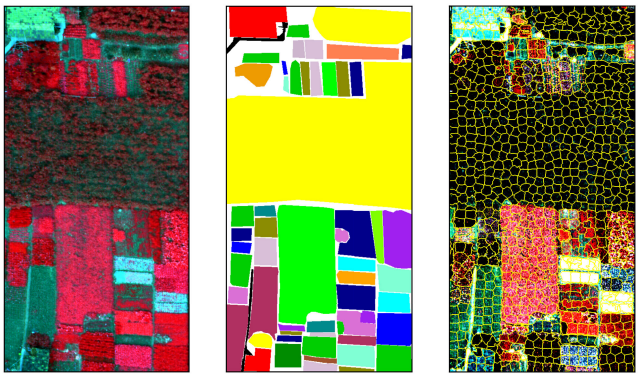
B. Experiment Settings

1) For Comparison Methods and Sample Selection: Recent state-of-the-art methods for HSI classification, compared

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TABLE III
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR SALINAS

No	Color	Name	Train.	Val.	Test.
1		Broccoli green weeds 1	5	1	2003
2		Broccoli green weeds 22	5	1	3720
3		Fallow	5	1	1970
4		Fallow rough plow	5	1	1388
5		Fallow smooth	5	1	2672
6		Stubble	5	1	3953
7		Celery	5	1	3573
8		Grapes untrained	5	1	11265
9		Soil vineyard develop	5	1	6197
10		Corn senesced green weeds	5	1	3272
11		Lettuce romaine 4wk	5	1	1062
12		Lettuce romaine 5wk	5	1	1921
13		Lettuce romaine 6wk	5	1	910
14		Lettuce romaine 7wk	5	1	1064
15		Vineyard untrained	5	1	7262
16		Vineyard vertical trellis	5	1	1801
		Total	80	16	54033



(a) (b) (c)

Fig. 6. WHU-Hi-HongHu. (a) False-color composite image. (b) Ground-truth map. (c) Segmentation result.

TABLE IV
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR WHU-HI-HONGHU

No	Color	Name	Train.	Val.	Test.
1		Red roof	5	1	14035
2		Road	5	1	3506
3		Bare soil	5	1	21815
4		Cotton	5	1	163279
5		Cotton firewood	5	1	6212
6		Rape	5	1	44551
7		Chinese cabbage	5	1	24097
8		Pakchoi	5	1	4048
9		Cabbage	5	1	10813
10		Tuber mustard	5	1	12388
11		Brassica parachinensis	5	1	11009
12		Brassica chinensis	5	1	8948
13		Small Brassica chinensis	5	1	22501
14		Lactuca sativa	5	1	7350
15		Celtuce	5	1	996
16		Film covered lettuce	5	1	7256
17		Romaine	5	1	3004
18		Carrot	5	1	3211
19		White radish	5	1	8706
20		Garlic sprout	5	1	3480
21		Broad bean	5	1	1322
22		Tree	5	1	4034
		Total	110	22	386561

with HGCN-MHF, include two traditional DL methods, i.e., 3D-CNN [54] and spectral–spatial residual network (SSRN) [13]; four popular FSL methods, i.e., deep cross-domain FSL (DCFSL) [55], spectral–spatial Siamese network (S3Net) [44], deep prototypical network with hybrid residual attention (DPN-HRA) [56], and self-pooling transformer (SPFormer) [22]; and four outstanding GCN-based methods, i.e., attention multihop graph and multiscale convolutional fusion network (AMGCFN) [39], CNN-enhanced graph convolutional network (CEGCN) [37], weighted feature fusion of CNN and graph attention network (WFCG) [38], and GCN and transformer fusion network (GTFN) [33]. For a fair comparison, all experiments are conducted using the hyperparameters recommended in their original paper. To evaluate the generalization performance, the labeled samples per class for training is set in the range of [5, 10, 15, 20, 25, 30]. If there are so few labeled samples for a certain class, the selected samples for training are required less than half of the total labeled samples. In this situation, half of the labeled samples for this class are used for training. Additionally, on average each class randomly selects one sample as the validation, and the other for testing.

2) For Evaluation Metrics: Five evaluation metrics are selected to assess the experimental results: overall accuracy (OA), average accuracy (AA), kappa statistics (Kappa), running time, and the number of network parameters (Parameters).

3) Running Platforms: All the experiments are implemented by PyTorch 1.13.0 in Python 3.8.16 environment and the hardware used for training is an Intel[1] Xeon[1] CPU E5-2683 v4 and a NVIDIA GeForce RTX 3090 GPU with 24-GB RAM. Note that all experiment results are reported by averaging the results after conducting ten independent runs in terms of randomly selected initial training samples. The source code will be available online: https://github.com/ZhaohuiXue/HGCN-MHF.

C. Hyperparameter Analysis

The training iterations (epochs) are set to 500, and the learning rate is set to 0.0005 for all datasets. Meanwhile, other parameters, including the different numerical values of SLIC's segmentation scale (λ), the diverse numbers of neighbor nodes in ε_{kNN} (κ), the depth of network (l), and adjustable parameters in two hyperedge weight functions (σ_1 and σ_2), are analyzed in this section.

1) Analysis of λ: To determine the optimal segmentation scale, λ is systematically varied within the range of [100, 200, 300, 400, 500]. Specifically, a larger scale produces a smaller graph, resulting in fewer superpixel nodes and larger superpixels, which can suppress noise. Conversely, a smaller scale produces a larger graph, consisting of small superpixels with noise. Fig. 7(a) illustrates the relationship between OA and the segmentation scale. When the segmentation scale λ is set to 100, 400, 300, and 500 for the four datasets, respectively, the highest OA can be achieved. The corresponding segmentation results are shown in Figs. 3(c), 4(c), 5(c), and 6(c).

2) Analysis of κ: To determine the optimal number of neighbor nodes, κ is varied within the range of [3, 6, 9, 12, 15, 18]. As shown in Fig. 7(b), the relationship between OA and the number of neighbor nodes is depicted. When the number of neighbor nodes is set to range from [3, 6, 9, 12] for the four

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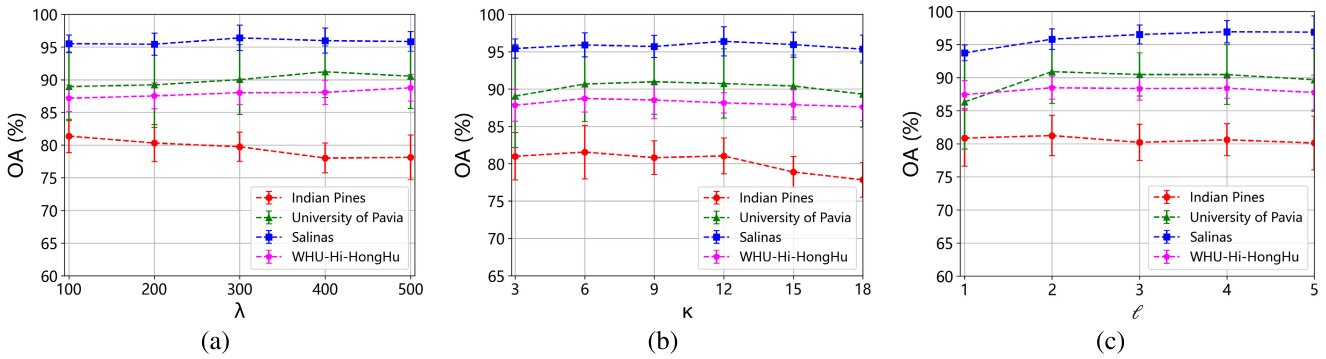


Fig. 7. Evaluation of OA as a function of (a) segmentation scale λ, (b) number of neighbor nodes κ, and (c) network depth l.

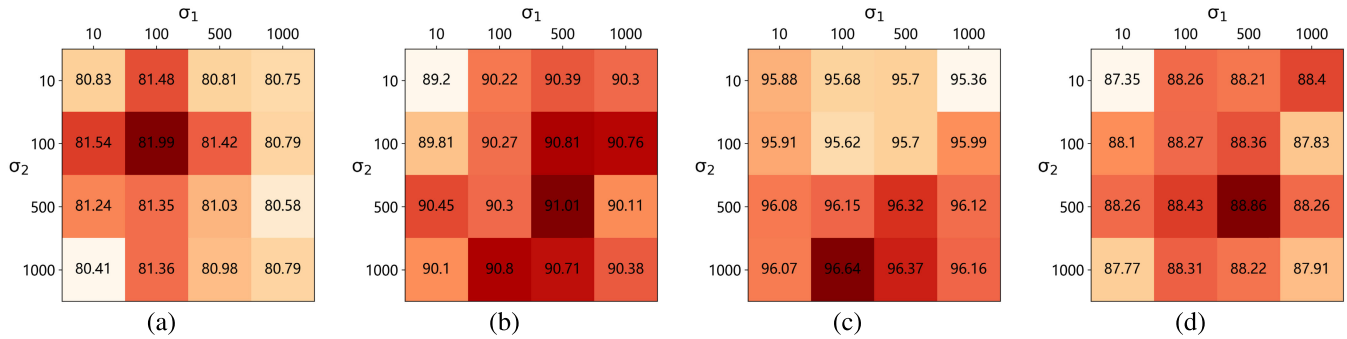


Fig. 8. Impact of differentiated σ_1 and σ_2 on OA for the four datasets. (a) Indian Pines. (b) University of Pavia. (c) Salinas. (d) WHU-Hi-HongHu.

datasets, the OA remains relatively insensitive and achieves a higher value. However, when the number of neighbor nodes exceeds 15, the OA begins to decrease slightly, but it does not decrease much lower than the optimal OA due to the fact that the nodes are selected in the 2-hop neighbors, which does not cover a large spatial range. Thus, κ is set to 6, 9, 12, and 6, respectively, for the four datasets to achieve the highest OA.

3) Analysis of l: To determine the optimal network depth, l is varied within the range of $[1, 2, 3, 4, 5]$. The depth of the network refers to the number of HGC layers, which significantly influences network performance. Generally, deeper networks yield smoother classification results. For datasets with large-scale regular objects, such as Salinas, this smoothing effect helps suppress noise and improve classification accuracy. Conversely, for datasets with small-scale objects, excessive smoothing can erase critical details and negatively impact performance. As illustrated in Fig. 7(c), the relationship between OA and network depth is shown. Consequently, setting the network depth l to 2, 2, 4, and 2 for the four datasets, respectively, achieves the highest OA.

4) Analysis of σ_1 and σ_2: To select the optimal spatial similarity measure factor and spectral similarity measure factor for determining the weight of hyperedges, σ_1 and σ_2 are set identically within the range of $[10, 100, 500, 1000]$. The evaluation of OA as a function of spatial and spectral similarity measure factor is illustrated in Fig. 8. When (σ_1, σ_2) is set to (100, 100), (500, 500), (100, 1000), and (500, 500) for the four datasets, respectively, the highest OA can be achieved. It should be noted that the σ_1 values for Indian Pines and Salinas are equal because they were acquired by the same sensor, while the σ_2 value for Indian Pines is smaller than that for Salinas due to the smaller image size of Indian Pines. The University of Pavia and WHU-Hi-HongHu were captured by different sensors, and their sizes are also different, so their σ_1 and σ_2 values are different.

D. Generalization Performance

To validate the generalization performance of HGCN-MHF, a few labeled samples are selected from each class, i.e., $[5, 10, 15, 20, 25, 30]$, to compare the classification results of different methods. Fig. 9(a)–(d) illustrates the evaluation of OA with different numbers of labeled training samples per class on four datasets. As depicted in the figures, HGCN-MHF almost achieves the highest accuracy when the training sample is limited for all datasets. In addition, our method gets the greater advantages in all cases for Indian Pines and Salinas datasets. Even with the increase in labeled samples, such as $[20, 25, 30]$, HGCN-MHF can still maintain the classification accuracy in the first echelon for University of Pavia and WHU-Hi-HongHu datasets. On the whole, the proposed HGCN-MHF algorithm outperforms the other methods in most cases, especially with small size of training samples. From this, the superiority of our method in terms of generalization performance can be proved.

E. Ablation Analysis

1) Effectiveness of MHF: To verify the effectiveness of the proposed MHF, we conducted a comparison of OA when using different hyperedges. MHF includes three kinds of hyperedge generation strategies, which are $\varepsilon_{\text{pair}}$, ε_{kNN}, and ε_{hop}. Table V states the averaged OA of individual hyperedge types and MHF, and the fusion of three groups of hyperedges proved to be the most optimal. It should be noted that fusing $\varepsilon_{\text{pair}}$

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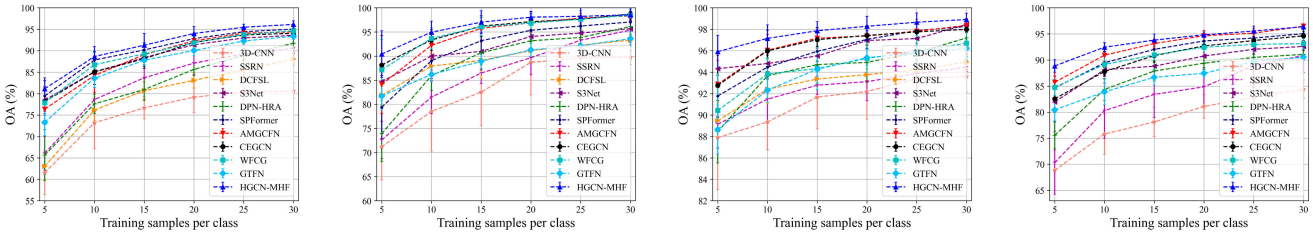


Fig. 9. Evaluation of OA as a function of number of labeled samples per class. (a) Indian Pines. (b) University of Pavia. (c) Salinas. (d) WHU-Hi-HongHu.

TABLE V

ABLATION ANALYSIS OF MHF IN FOUR DATASETS

ε_{kNN}	ε_{pair}	ε_{hop}	Indian Pines	University of Pavia	Salinas	WHU-Hi-HongHu
✓	×	×	78.54±3.85	89.48±4.73	95.22±1.81	87.15±1.48
×	✓	×	81.18±2.83	90.13±5.12	95.53±1.50	88.00±1.98
×	×	✓	79.71±2.43	89.80±6.06	95.78±1.67	87.31±2.38
✓	✓	×	81.32±3.12	90.74±4.31	95.71±1.88	**89.07±1.12**
✓	×	✓	80.53±2.56	90.62±4.21	95.80±1.31	88.84±2.18
×	✓	✓	80.58±3.52	89.79±4.60	95.66±1.50	88.20±2.27
✓	✓	✓	**81.99±2.22**	**91.01±4.31**	**96.64±1.54**	88.86±1.19

Note: ε_{kNN} denotes hyperedges connecting the central node with its K nearest neighbours nodes of spectral feature in 2-hop spatial neighbours; ε_{pair} denotes hyperedges connecting the central node with its neighbours in spatial space; ε_{hop} denotes hyperedges connecting the central node with its 1-hop neighboring nodes.

TABLE VI

ABLATION ANALYSIS OF THE WEIGHT FUNCTION IN FOUR DATASETS

Methods	Indian Pines	University of Pavia	Salinas	WHU-Hi-HongHu
$W = 1$	80.97±3.08	89.07±5.84	95.53±1.10	86.71±1.57
$W_{spatial}$	81.17±2.19	89.60±4.71	95.65±1.52	88.45±1.10
$W_{spectral}$	81.51±3.14	90.59±4.48	95.92±1.75	88.22±1.67
$W_{spe-spa}$	**81.99±2.22**	**91.01±4.31**	**96.64±1.54**	**88.86±1.19**

and ε_{hop} is not necessarily more effective than using a single hyperedge generation strategy. This is because both $\varepsilon_{\text{pair}}$ and ε_{hop} are based on the topological adjacency structure, resulting in the use of the same nodes in HGC feature extraction, which may lead to overfitting and produce only marginal improvements. Additionally, fusing ε_{kNN}, based on spectral feature similarity, with either $\varepsilon_{\text{pair}}$ or ε_{hop}, based on spatial adjacency, yields better results. Overall, the fusion of all three strategies achieves the better OA.

Additionally, to validate the performance improvement of the weight function, we compared four variants of the weight function. The first variant, $W = 1$, assigns equal weights of 1 to all hyperedges without distinction. The second variant, W_{spatial}, utilizes spatial Euclidean distance as the metric function for determining hyperedge weights. The third variant, W_{spectral}, employs spectral distance as the metric function for defining hyperedge weights. The fourth variant, $W_{\text{spe-spa}}$, integrates both spatial and spectral distances as the metric function for confirming hyperedge weights, representing our proposed method. Table VI demonstrates that the averaged OA of $W_{\text{spe-spa}}$, which is our proposed method for determining weights of hyperedges, consistently outperforms the other variants across all datasets.

2) Effectiveness of SWFF: To examine the performance improvement of the proposed SWFF, five variants of HGCN-MHF were compared. The two basic variants involve using only CNN branch or HGCN branch as the model. Another variant involves concatenating CNN branch and HGCN branch, referred to as S_{concat}. The remaining two

TABLE VII

ABLATION ANALYSIS OF SWFF IN FOUR DATASETS

Methods	Indian Pines	University of Pavia	Salinas	WHU-Hi-HongHu
$CNNBranch$	75.44±2.77	79.04±8.08	90.43±1.48	82.33±2.74
$HGCNBranch$	78.11±3.28	85.07±5.34	96.35±1.03	86.60±2.27
S_{concat}	79.40±3.35	89.32±5.75	95.64±1.22	87.51±1.48
S_{score_1}	79.17±3.84	87.80±5.00	96.12±1.77	87.07±2.35
S_{score_2}	**81.99±2.22**	**91.01±4.31**	**96.64±1.54**	**88.86±1.19**

Note: $CNNBranch$ refers to classification using only CNN feature; $HGCNBranch$ refers to classification using only HGCN feature; S_{concat} refers to classification using concatenated feature of CNN feature and HGCN feature; S_{score_1} refers to classification using score weighted feature of CNN feature and HGCN feature; S_{score_2} refers to classification using score weighted feature of CNN feature, HGCN feature and their concatenated feature.

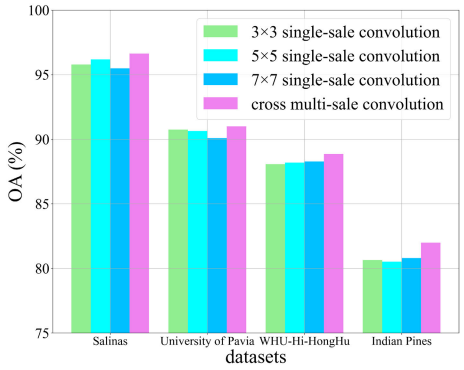


Fig. 10. Ablation analysis of CMC in four datasets.

variants employ score-weighted strategies, one of them employs the score-weighted feature of CNN branch and HGCN branch, denoted as S_{score_1}. And the other is the score-weighted feature of CNN branch, HGCN branch, and their concatenated feature, denoted as S_{score_2}. Table VII states the averaged OA of S_{score_2}, which is our proposed method and achieves higher than others in all datasets. It is noteworthy that HGCN branch has excellent performance on the Salinas dataset, since Salinas is neatly grouped and the segmentation results fit the classification results of ground objects. However, other datasets have no good segmentation results because of the irregular objects distribution. In this situation, CNN branch needs to compensate for the shortcomings of HGCN branch. Among the three feature fusion methods, S_{score_2}(ours) achieves the best performance.

3) Effectiveness of CMC: To examine the effectiveness of the proposed CMC, we compared four variants of HGCN-MHF: HGCN-MHF with only 3×3 single-scale convolution, HGCN-MHF with only 5×5 single-scale convolution, HGCN-MHF with only 7×7 single-scale convolution, and HGCN-MHF with CMC. As shown in Fig. 10, the model with CMC consistently achieves higher OA than those with

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TABLE VIII

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR INDIAN PINES (USING FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3D-CNN [54]	SSRN [13]	DCFSL [55]	S3Net [44]	DPN-HRA [56]	SPFormer [22]	AMGCFN [39]	CEGCN [37]	WFCG [38]	GTFN [33]	HGCN-MHF
1	78.17±18.03	80.30±16.22	97.32±3.53	**99.27±2.20**	34.57±15.76	98.78±1.21	98.05±1.82	96.83±2.45	97.56±1.54	97.31±2.77	99.02±1.19
2	52.30±20.42	52.95±11.59	40.44±7.11	62.35±8.34	62.53±12.27	60.92±13.72	57.04±8.49	57.13±12.47	62.77±11.91	56.30±14.72	**64.43±14.96**
3	57.97±22.90	53.64±12.12	45.62±11.09	62.41±13.24	56.22±9.87	63.73±10.85	72.59±16.06	64.17±13.78	67.04±15.78	52.30±10.40	**74.38±14.62**
4	62.15±23.24	70.69±14.78	78.36±9.35	**98.62±1.70**	40.44±11.35	96.29±4.79	75.70±16.12	89.48±10.37	97.45±3.89	89.05±8.03	93.05±6.95
5	73.76±13.44	**85.07±12.77**	73.08±6.40	78.24±7.81	80.30±17.75	81.96±4.41	76.35±7.11	77.61±7.35	79.25±8.27	80.64±5.83	77.99±8.00
6	78.29±8.74	89.00±7.04	82.46±6.53	94.88±2.80	92.87±6.72	95.26±2.99	86.11±17.39	96.76±2.20	**97.57±2.36**	92.17±5.12	96.39±2.77
7	76.84±20.5	81.38±20.77	**100.00±0.00**	**100.00±0.00**	32.34±17.27	99.56±1.30	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.56±1.30	**100.00±0.00**
8	85.60±12.89	76.34±16.16	81.44±13.65	99.94±0.19	97.28±2.90	99.95±0.12	98.87±2.69	98.66±3.52	99.41±1.78	96.06±3.11	**99.98±0.06**
9	75.55±15.41	66.58±19.74	98.67±2.67	**100.00±0.00**	16.17±13.75	99.33±2.00	**100.00±0.00**	96.67±6.83	**100.00±0.00**	99.33±2.00	**100.00±0.00**
10	59.54±23.01	61.34±12.14	59.06±7.05	**74.78±12.2**	56.05±13.86	71.69±7.83	69.39±12.70	69.50±11.91	68.38±9.01	65.52±7.62	73.12±8.45
11	48.60±20.29	54.62±16.59	59.23±12.48	73.29±8.80	80.40±6.27	76.28±7.62	76.58±11.08	81.18±9.92	74.10±9.77	69.04±8.37	**81.82±9.06**
12	52.82±16.91	64.04±15.08	44.15±10.39	72.43±8.82	40.32±15.38	79.57±13.11	64.89±11.58	57.46±18.31	70.63±12.17	70.44±10.12	**74.98±11.99**
13	85.37±9.43	87.30±20.27	99.1±0.97	98.25±2.95	83.21±14.71	98.65±16.43	99.25±1.54	98.85±1.14	99.10±1.81	96.25±3.17	**99.60±0.53**
14	83.74±9.90	87.00±8.24	81.48±12.04	92.17±8.24	**94.29±4.86**	94.15±4.94	85.95±12.60	90.12±10.87	85.85±11.84	86.24±11.81	91.37±9.03
15	59.31±13.53	75.53±14.54	70.37±12.57	**95.17±4.88**	68.78±14.30	85.93±12.49	91.10±12.68	94.51±8.97	92.10±9.64	84.96±10.06	95.01±7.68
16	92.73±2.52	91.04±6.02	98.64±2.58	98.41±3.34	66.10±21.70	99.54±1.36	99.66±0.73	**99.77±0.68**	99.32±1.70	97.38±1.76	99.43±1.16
OA (%)	61.52±5.06	66.06±3.91	62.99±2.99	78.49±3.17	65.63±5.97	79.44±3.55	76.37±3.18	78.14±3.22	77.85±2.85	73.29±3.10	**81.99±2.22**
AA (%)	70.17±2.58	73.55±5.56	75.59±1.11	87.42±1.58	65.97±3.73	87.60±1.67	84.47±2.38	85.54±1.89	86.91±1.78	83.28±1.29	**88.89±1.17**
$\kappa\times100$	57.73±5.68	62.96±4.18	58.37±3.01	75.82±3.45	61.80±6.17	76.81±3.95	73.26±3.55	75.19±3.63	75.17±3.17	69.98±3.37	**79.58±2.51**

TABLE IX

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR UNIVERSITY OF PAVIA (USING FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3D-CNN [54]	SSRN [13]	DCFSL [55]	S3Net [44]	DPN-HRA [56]	SPFormer [22]	AMGCFN [39]	CEGCN [37]	WFCG [38]	GTFN [33]	HGCN-MHF
1	71.60±18.07	81.09±7.31	77.89±12.85	85.99±7.48	78.26±27.98	83.96±8.48	74.11±9.57	86.90±5.18	87.98±5.82	75.98±6.82	**93.88±4.06**
2	64.66±26.45	66.29±12.45	87.38±3.86	81.82±14.35	85.02±3.58	64.99±14.92	82.08±10.80	82.03±13.61	81.26±13.04	80.67±6.24	**85.68±9.90**
3	59.54±21.08	64.12±13.42	65.53±12.97	82.82±7.02	53.31±15.91	81.11±10.06	89.14±15.89	**98.07±3.51**	94.04±7.24	75.92±7.72	93.93±9.20
4	79.98±25.88	85.72±13.21	**92.50±4.36**	89.41±4.48	74.66±13.42	93.56±4.51	78.82±11.46	89.36±3.65	90.87±3.23	82.99±6.55	90.81±4.94
5	83.99±18.31	99.41±0.57	99.31±0.98	99.27±1.07	99.28±1.14	**100.00±0.00**	99.06±0.68	99.86±0.19	99.86±0.28	99.31±0.60	99.94±0.13
6	69.40±17.94	71.95±16.75	74.07±5.47	84.79±10.31	65.71±14.39	95.83±6.83	93.27±12.97	95.31±10.17	94.92±6.57	90.77±5.56	**97.98±4.85**
7	61.59±27.06	73.27±15.42	78.94±7.90	99.77±0.40	44.71±24.32	99.84±0.17	99.23±1.22	97.99±4.72	97.81±3.94	98.20±1.51	**99.88±0.10**
8	40.82±22.72	73.97±10.40	65.53±11.79	84.29±18.06	54.99±20.37	89.00±6.96	88.93±14.30	**95.86±3.49**	87.72±8.08	71.89±9.98	94.66±10.01
9	93.94±7.39	99.08±1.12	**99.11±0.76**	82.24±10.76	75.84±14.85	98.98±0.79	91.94±8.25	93.59±6.25	94.19±7.77	94.30±4.03	93.57±11.73
OA (%)	71.18±6.85	72.60±3.88	81.86±1.91	84.73±5.57	73.89±5.78	79.39±6.59	84.15±5.98	88.16±6.09	87.16±6.35	81.73±2.88	**91.01±4.31**
AA (%)	71.91±8.37	79.43±4.03	81.81±1.47	87.82±2.95	71.31±6.04	89.70±1.83	88.51±3.35	93.23±2.12	92.07±3.22	85.56±1.28	**94.48±2.29**
$\kappa\times100$	65.23±8.32	67.58±4.62	76.41±2.31	80.56±6.45	67.00±6.49	74.70±7.41	79.78±7.20	85.03±7.28	83.76±7.60	76.69±3.39	**88.50±5.28**

single-scale convolutions, demonstrating the effectiveness of the proposed CMC block. Specifically, the classification accuracy of the model without CMC decreased by 1.19%–1.47%, 0.26%–0.92%, 0.45%–1.15%, and 0.57%–0.78% in Indian Pines, University of Pavia, Salinas, and WHI-Hi-HongHu, respectively.

F. Classification Results

In this part, we randomly select five labeled samples from each class for training to compare the performance of 3D-CNN [54], SSRN [13], DCFSL [55], S3Net [44], DPN-HRA [56], SPFormer [22], AMGCFN [39], CEGCN [37], WFCG [38], GTFN [33], and our proposed HGCN-MHF. It is noted that DCFSL can only distinguish 19 classes at most, so this model cannot be evaluated by the WHU-Hi-HongHu dataset (22 classes).

1) Indian Pines: Due to the low spatial resolution of Indian Pines dataset, there are some problems such as pixel confusion, boundary ambiguity, and detail information loss, resulting in low classification accuracy compared with other datasets. According to the results reported in Table VIII, the accuracies of HGCN-MHF are significantly higher than those of the other methods. Specifically, HGCN-MHF achieves the highest OA of 81.99%, representing an increase from 3.50% to 20.47% compared to other methods. Additionally, the AA and κ of HGCN-MHF are also noticeably improved. As shown in Fig. 11, the classification map of HGCN-MHF is evidently less noisy and smoother. Moreover, HGCN-MHF attains the highest accuracy in eight of 16 classes, further highlighting the superiority of our method.

2) University of Pavia: Although University of Pavia dataset has high spatial resolution, the complex and irregular distribution of ground objects makes it difficult to classify, resulting in medium classification accuracy compared with other datasets. According to the results reported in Table IX, the accuracies of HGCN-MHF are significantly higher than those of the other methods. Specifically, HGCN-MHF achieves the highest OA of 91.01%, representing an increase from 2.85% to 19.83% compared to other methods. Besides, the AA and κ of HGCN-MHF are also notably improved. As shown in Fig. 12, the classification map of HGCN-MHF is evidently less noisy and smoother. Moreover, HGCN-MHF attains the highest accuracies in four of nine classes, demonstrating the superiority of HGCN-MHF.

3) Salinas: Due to the high spatial resolution of Salinas dataset, the smooth and regular distribution of ground objects makes the classification simple, resulting in high classification accuracy than other datasets. According to the results reported in Table X, the accuracies of HGCN-MHF are significantly higher than those of the other methods. Specifically, HGCN-MHF achieves the highest OA of 96.64%, representing an increase from 2.31% to 8.75% compared to other methods. Besides, the AA and κ of HGCN-MHF are also noticeably improved. As shown in Fig. 13, the classification map of HGCN-MHF is evidently less noisy and smoother. Moreover, HGCN-MHF attains the highest accuracies in eight of 16 classes, demonstrating the superiority of HGCN-MHF.

4) WHU-Hi-HongHu: Although WHU-Hi-HongHu dataset has the highest spatial resolution and regular spatial distribution of ground objects, the spectral characteristics of

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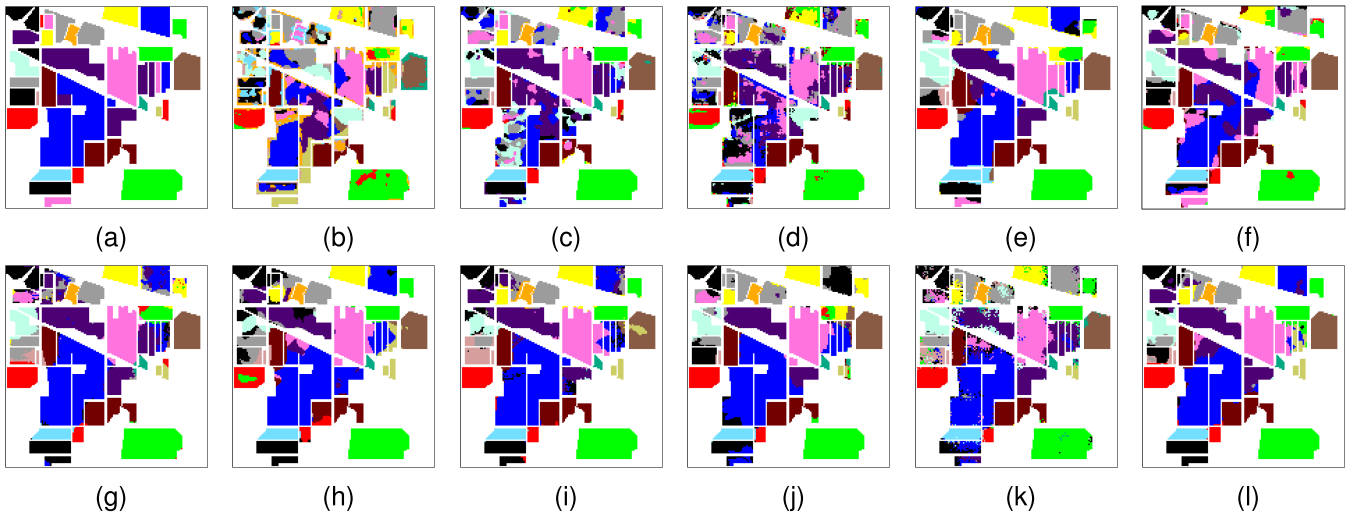


Fig. 11. Classification maps obtained by different methods for Indian Pines. (a) Ground truth. (b) 3D-CNN (61.52%). (c) SSRN (66.06%). (d) DCFSL (62.99%). (e) S3Net (78.49%). (f) DPN-HRA (65.63%). (g) SPFormer (79.44%). (h) AMGCFN (76.37%). (i) CEGCN (78.14%). (j) WFCG (77.85%). (k) GTFN (73.29%). (l) HGCN-MHF (81.99%).

TABLE X

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR SALINAS (USING FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3D-CNN [54]	SSRN [13]	DCFSL [55]	S3Net [44]	DPN-HRA [56]	SPFormer [22]	AMGCFN [39]	CEGCN [37]	WFCG [38]	GTFN [33]	HGCN-MHF
1	97.56±3.49	98.81±1.53	98.99±1.12	99.65±0.46	98.28±4.19	99.90±0.25	**100.00±0.00**	99.82±0.43	99.48±1.03	97.59±2.88	99.19±2.10
2	96.94±2.06	99.28±1.04	99.59±0.92	99.44±1.12	97.66±2.06	**100.00±0.00**	99.96±0.03	99.91±0.14	99.50±1.27	94.78±7.94	**100.00±0.00**
3	79.68±20.61	87.65±8.11	90.44±10.53	99.90±0.26	98.02±2.28	92.24±11.30	**100.00±0.00**	**100.00±0.00**	99.11±2.08	97.89±4.17	**100.00±0.00**
4	95.30±4.63	93.13±5.65	99.48±0.62	99.92±0.09	92.96±5.07	**99.97±0.04**	99.38±1.01	99.87±0.11	99.77±0.19	96.11±2.90	99.94±0.10
5	93.62±3.50	94.29±4.32	91.08±3.03	96.06±3.56	95.34±2.05	96.54±3.87	93.83±5.68	**97.85±1.62**	97.31±1.87	94.62±2.36	97.45±1.45
6	99.49±0.62	99.41±1.38	98.44±2.20	97.65±3.80	99.79±0.44	99.66±0.47	99.91±0.09	99.69±0.23	**99.85±0.18**	97.71±1.56	99.69±0.29
7	95.36±2.62	99.32±0.61	98.98±1.77	99.11±1.63	97.82±2.34	99.78±2.89	99.88±0.26	99.90±0.08	99.99±0.03	97.64±2.20	**99.99±0.02**
8	87.75±6.37	83.38±9.95	77.86±9.76	86.11±5.78	85.83±9.10	78.23±11.58	78.18±13.16	77.58±14.20	75.04±20.35	74.32±7.70	**92.53±5.03**
9	95.01±10.75	95.27±4.83	99.07±1.26	98.78±2.67	98.12±1.93	99.62±0.60	**100.00±0.00**	**100.00±0.00**	99.64±1.06	99.51±0.46	**100.00±0.00**
10	80.11±13.27	86.59±12.08	87.02±3.49	90.19±7.06	95.19±4.03	92.18±4.57	92.05±6.37	89.01±7.31	**92.29±5.88**	86.10±11.36	91.48±7.54
11	82.35±14.47	88.08±9.23	98.57±1.36	99.88±0.25	94.29±7.92	99.86±0.08	97.86±2.44	99.12±1.59	99.43±1.03	98.06±2.43	**99.90±0.18**
12	78.63±14.92	93.37±6.18	99.61±0.56	95.98±2.88	89.18±8.25	99.87±0.18	99.79±0.34	99.98±0.03	99.23±2.09	93.82±6.00	**99.96±0.09**
13	98.06±1.61	83.42±17.74	99.25±0.34	97.99±2.49	92.88±7.82	98.58±1.22	99.01±1.38	**99.75±0.50**	99.19±1.30	92.70±8.28	99.59±0.53
14	78.83±13.73	83.79±11.62	**98.55±1.73**	91.76±7.99	90.06±6.56	97.24±0.47	96.19±5.49	97.22±2.05	95.85±3.43	90.99±4.91	98.13±2.90
15	75.60±22.31	77.71±12.20	73.04±6.60	91.27±6.51	66.47±11.67	80.85±7.77	87.97±11.55	87.57±16.64	74.11±26.03	75.87±10.83	**92.35±7.80**
16	90.40±7.57	94.54±8.20	91.71±6.05	98.67±1.13	97.67±3.64	99.62±0.63	99.84±0.46	**100.00±0.00**	99.56±1.29	97.49±2.12	99.14±1.73
OA (%)	87.89±4.84	89.14±2.25	89.46±1.68	94.33±1.22	88.44±2.92	91.78±1.94	92.87±2.22	92.76±3.01	90.42±3.44	88.65±2.28	**96.64±1.54**
AA (%)	89.04±2.91	91.13±2.43	93.86±0.76	96.40±0.79	93.10±1.37	95.89±0.89	96.49±1.02	96.71±1.27	95.58±1.25	92.83±1.61	**98.08±0.67**
$\kappa\times100$	86.62±5.29	87.97±2.47	88.29±1.84	93.70±1.35	87.19±3.21	90.87±2.13	92.08±2.45	91.97±3.32	89.38±3.80	87.40±2.52	**96.26±1.71**

TABLE XI

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR WHU-HI-HONGHU (USING FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3D-CNN [54]	SSRN [13]	S3Net [44]	DPN-HRA [56]	SPFormer [22]	AMGCFN [39]	CEGCN [37]	WFCG [38]	GTFN [33]	HGCN-MHF
1	84.96±13.12	81.78±11.38	80.92±9.44	**90.95±7.33**	85.24±9.27	81.09±13.51	81.19±12.91	82.87±9.93	87.54±3.15	84.31±12.26
2	75.99±11.66	65.73±21.98	68.69±7.91	54.83±19.16	84.86±10.30	77.35±18.09	79.69±7.33	80.43±17.42	74.48±6.69	**85.12±14.31**
3	84.10±12.30	89.66±7.54	84.68±4.67	93.29±5.21	81.91±3.68	85.89±2.93	**90.92±3.63**	88.84±2.77	77.95±2.70	88.30±4.43
4	65.62±13.04	72.86±9.28	88.67±7.85	**98.09±1.15**	89.23±5.70	92.82±3.94	88.19±5.17	92.23±2.49	87.37±3.68	95.69±1.96
5	69.93±13.15	67.50±15.92	90.07±6.67	30.84±10.85	90.17±4.84	94.26±7.52	**96.39±5.89**	95.26±3.69	84.44±4.77	91.98±4.11
6	63.36±22.28	88.70±5.38	92.39±2.82	90.92±3.21	88.23±3.57	**94.28±3.76**	92.70±2.30	91.54±4.20	85.99±2.86	93.86±4.14
7	80.30±11.44	75.95±8.31	46.94±18.60	**87.39±1.95**	67.14±6.54	66.01±12.42	58.67±5.00	64.68±9.31	56.60±12.72	73.72±8.45
8	62.62±17.30	48.35±17.05	65.21±12.25	33.42±15.40	61.19±6.45	**73.12±11.00**	61.92±14.12	59.08±17.64	64.53±7.40	70.72±15.99
9	91.13±7.04	92.04±6.84	**95.63±4.67**	90.73±7.59	93.57±4.44	91.84±4.82	94.36±1.61	93.92±1.38	91.90±4.00	94.10±2.21
10	50.99±20.58	63.77±18.92	66.95±15.48	75.19±11.53	**88.52±5.36**	82.24±9.58	62.28±15.80	77.11±9.00	69.30±6.56	85.65±7.13
11	64.62±15.00	49.91±13.47	55.02±18.84	65.13±13.16	**73.32±10.85**	64.47±12.64	48.22±10.39	58.47±11.63	66.26±7.26	65.27±11.75
12	63.87±17.72	61.25±9.00	62.04±8.21	43.01±8.65	66.93±15.33	72.30±10.93	**77.15±5.46**	71.49±9.53	56.45±9.20	74.11±8.79
13	60.91±10.99	62.06±9.19	63.46±9.73	66.72±9.06	65.24±5.73	62.20±14.31	62.52±8.16	62.75±7.95	55.86±9.15	**67.58±5.30**
14	78.11±10.63	76.54±3.98	77.80±8.50	60.00±24.79	**86.53±11.26**	73.41±12.42	61.83±13.83	67.95±23.84	72.34±6.33	81.57±12.11
15	63.23±16.05	48.01±28.19	98.24±1.13	53.64±29.80	97.76±2.07	97.51±2.07	97.89±2.28	96.49±2.77	94.91±1.77	**98.56±1.93**
16	76.17±14.13	92.97±5.03	93.63±4.05	88.86±11.55	93.61±4.31	78.81±19.22	89.42±7.63	86.97±11.19	91.50±3.11	**93.68±5.24**
17	74.52±20.05	80.39±10.72	89.13±12.47	66.55±11.51	79.77±14.24	80.04±12.36	84.53±9.16	78.96±15.16	79.92±10.00	**93.82±9.17**
18	69.77±10.91	61.22±22.06	94.99±4.54	38.82±23.66	94.47±3.66	95.35±5.63	**97.99±2.47**	88.14±9.30	91.67±5.64	96.37±4.41
19	60.00±19.69	71.91±19.12	86.74±8.00	75.56±10.14	**87.93±6.27**	78.23±9.68	80.20±7.75	78.16±9.56	83.07±3.97	87.10±7.52
20	54.18±11.99	61.77±26.45	83.20±13.14	46.97±9.08	**89.83±16.73**	88.33±19.62	86.37±18.37	72.03±22.41	85.64±4.50	87.81±19.98
21	73.01±15.87	57.26±22.48	98.55±1.35	22.58±3.85	97.17±3.68	99.65±1.04	99.41±1.11	94.52±10.39	97.13±1.92	**99.72±0.52**
22	73.78±15.71	66.91±11.68	90.44±7.79	25.12±9.57	90.80±10.90	95.36±7.58	**98.95±2.04**	95.78±5.91	84.54±6.73	95.94±3.43
OA (%)	68.83±3.86	70.29±6.30	81.97±3.82	75.54±2.58	84.64±2.27	85.73±2.82	82.56±2.59	84.76±1.52	80.44±2.11	**88.86±1.19**
AA (%)	70.19±3.56	69.84±6.16	80.61±3.61	63.68±1.67	84.25±1.11	82.94±2.44	81.41±2.07	80.80±1.83	79.06±0.73	**86.14±1.64**
$\kappa\times100$	66.18±3.73	66.79±5.79	77.81±4.29	70.35±2.83	81.11±2.56	82.27±3.37	78.58±2.94	81.11±1.80	75.96±2.45	**86.06±1.46**

different ground objects are very similar and difficult to distinguish, such as cabbage and tuber mustard, resulting in the classification accuracy relative to other datasets. According to the results reported in Table XI, the accuracies of HGCN-MHF

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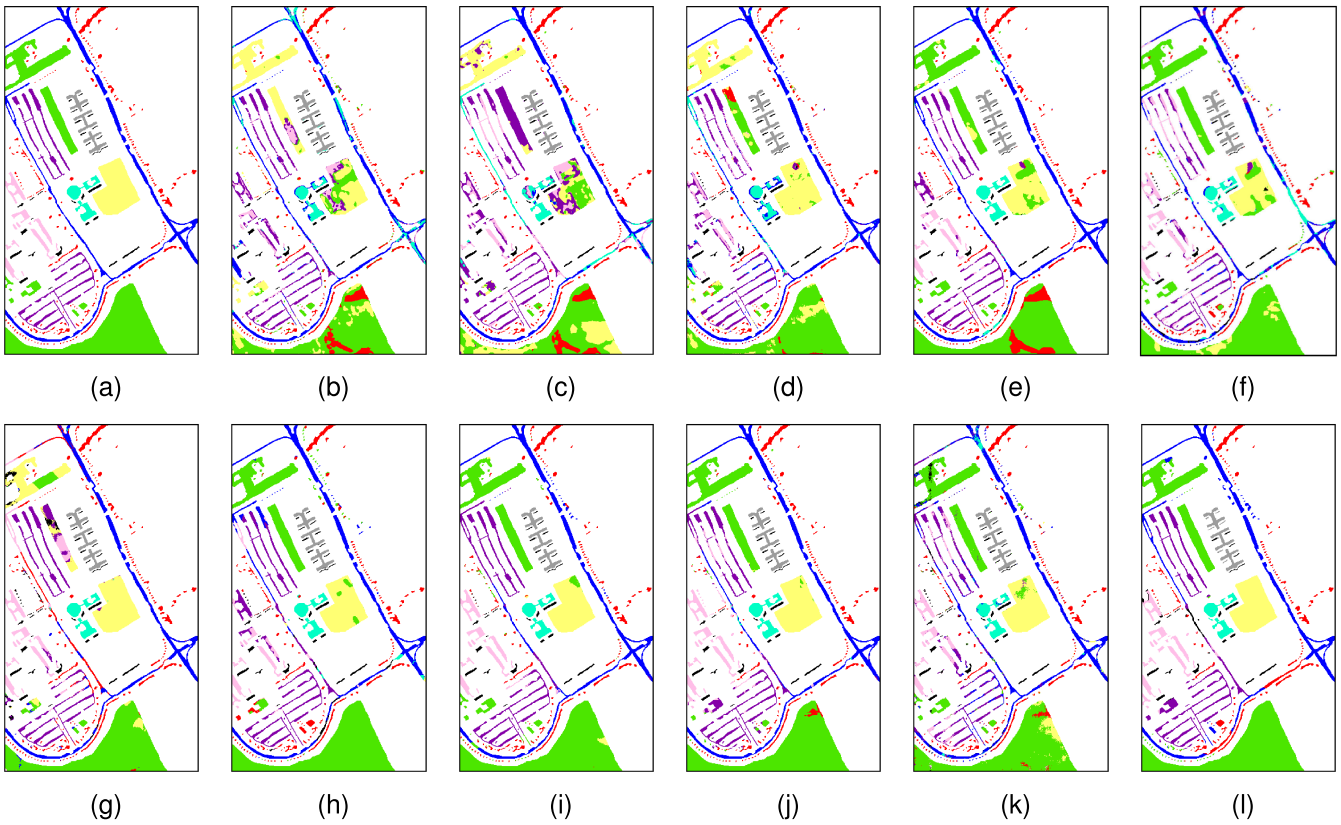


Fig. 12. Classification maps obtained by different methods for University of Pavia. (a) Ground truth. (b) 3D-CNN (71.18%). (c) SSRN (72.60%). (d) DCFSL (81.86%). (e) S3Net (84.73%). (f) DPN-HRA (73.89%). (g) SPFormer (79.39%). (h) AMGCFN (84.15%). (i) CEGCN (88.16%). (j) WFCG (87.16%). (k) GTFN (81.73%). (l) HGCN-MHF (91.01%).

TABLE XII
EVALUATION OF OA AS A FUNCTION OF DIFFERENCE kNN HYPEREDGE GROUP

data set	kNN in 1-hop	kNN in 2-hop	kNN in 3-hop	kNN in full-graph
Indian Pines	81.58±2.03	**81.99±2.22**	80.55±2.72	76.80±3.63
University of Pavia	90.83±3.32	**91.01±4.31**	88.62±5.79	83.33±7.48
Salinas	96.44±1.61	**96.64±1.54**	95.39±1.93	92.65±1.33
WHU-Hi-HongHu	88.41±1.72	**88.86±1.19**	87.67±1.98	86.83±1.31

TABLE XIII
EVALUATION OF OA AS A FUNCTION OF DIFFERENCE HGNN

Methods	Indian Pines	University of Pavia	Salinas	WHU-Hi-HongHu
HGNNConv	81.13±2.94	90.67±4.37	96.11±1.08	88.25±1.37
HGNNConv$^+$	**81.99±2.22**	**91.01±4.31**	**96.64±1.54**	**88.86±1.19**

are significantly higher than those of the other methods. Specifically, HGCN-MHF achieves the OA of 88.86%, representing an increase from 3.13% to 20.03% compared to other methods. Besides, the AA and κ of HGCN-MHF are also noticeably improved. As shown in Fig. 14, the classification map of HGCN-MHF is evidently less noisy and smoother. Moreover, HGCN-MHF attains the highest accuracies in six of 22 classes, demonstrating the superiority of HGCN-MHF.

IV. DISCUSSION

A. Analysis of kNN in Different Hops

To evaluate the effectiveness of selecting nodes within the 2-hop neighborhood, where their features closely resemble those of the central node, we compared four variants of ε_{kNN}: kNN in 1-hop, kNN in 2-hop, kNN in 3-hop, and kNN in the full graph. The distinction between these variants lies in the selection of nodes from different spatial regions, either within the k-hop neighborhood or across the entire graph. As observed in Table XII, hyperedges employing kNN in the 2-hop neighborhood achieve superior OA. Furthermore, hyperedges utilizing kNN in the 1-hop neighborhood also exhibit excellent OA. However, such hyperedges may introduce redundancy with ε_{hop} when the value of κ becomes large. Therefore, the ε_{kNN} adopting kNN in 2-hop is selected as one of the hypergraph generation modules in our approach.

B. Analysis of HGNNConv$^+$

Spatial domain HGC and frequency domain HGC are two distinct convolutional operations in HGNN. Spatial domain HGC is performed between nodes and their neighbors, where the information of neighbors is aggregated to the current node using a weight matrix, such as HGNNConv$^+$ [52]. Frequency domain HGC is a convolutional operation applied after transforming the hypergraph signal to the frequency domain, as demonstrated in HGNNConv [46]. To assess the effectiveness of two methods, they are both incorporated into our model, and OA is used as the evaluation criterion. Table XIII indicates that the OA of HGNNConv$^+$ is increased by 0.86%, 0.34%, 0.53%, and 0.61% on four datasets compared to the OA of HGNNConv. HGNNConv$^+$ is originally an advanced

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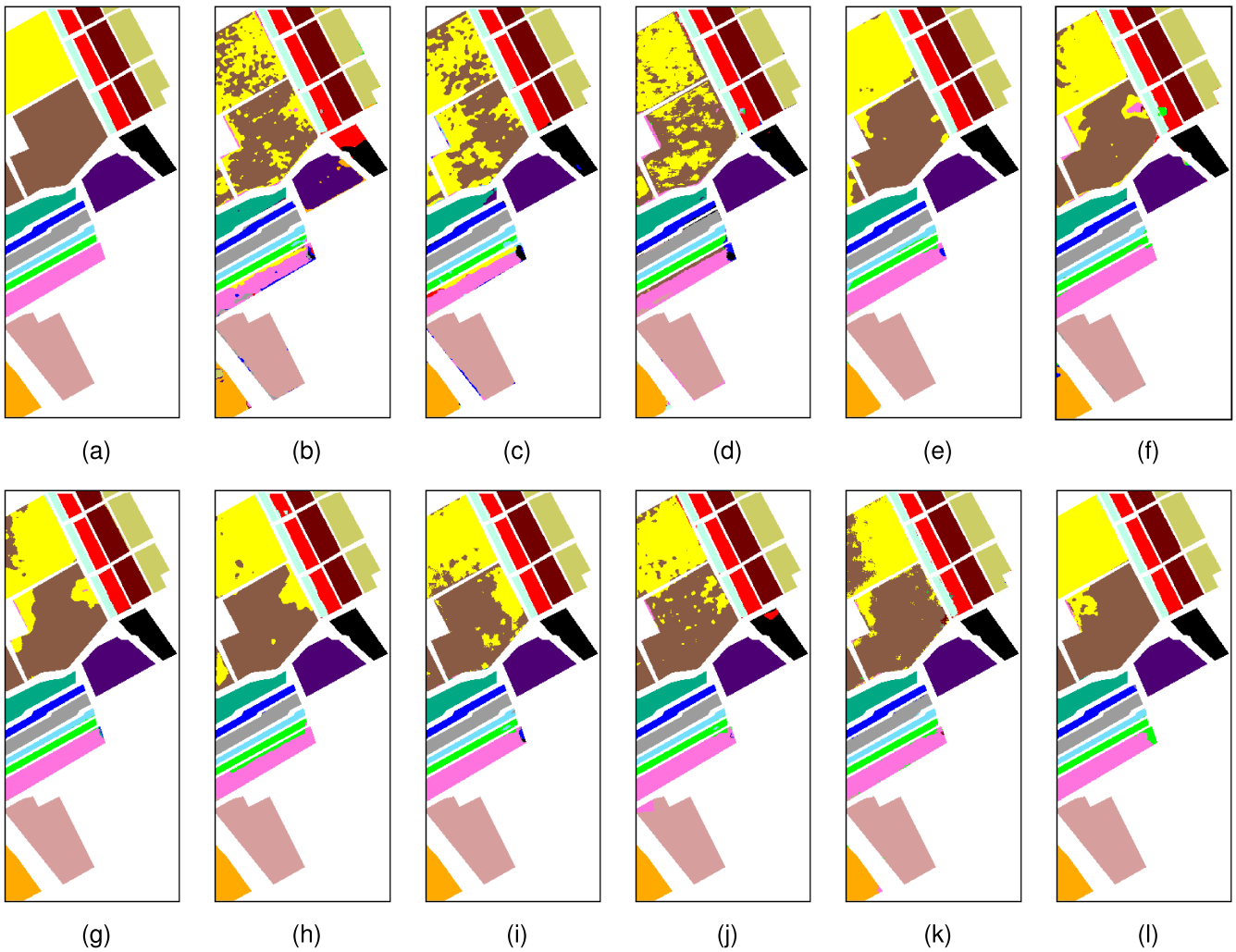


Fig. 13. Classification maps obtained by different methods for Salinas. (a) Ground truth. (b) 3D-CNN (87.89%). (c) SSRN (87.97%). (d) DCFSL (89.46%). (e) S3Net (94.33%). (f) DPN-HRA (88.44%). (g) SPFormer (91.78%). (h) AMGCFN (92.87%). (i) CEGCN (92.76%). (j) WFCG (90.42%). (k) GTFN (88.65%). (l) HGCN-MHF (96.64%).

TABLE XIV
RUNNING TIME AND PARAMETER SIZE OF DIFFERENT METHODS

Class	Time	3D-CNN	SSRN	DCFSL	S3Net	DPN-HRA	SPFormer	AMGCFN	CEGCN	WFCG	GTFN	HGCN-MHF(ours)
Indian Pines	Train(s)	3.72	4.85	1294.79	4.07	69.73	10.86	32.09	14.80	26.47	55.91	15.27
	Test(s)	1.93	0.93	1.46	5.21	46.77	1.75	0.02	1.41	1.29	10.50	1.75
	Parameters	313,964	540,808	4268,994	155,008	2636,997	86,296	368,217	166,370	102,422	281,878	174,546
University of Pavia	Train(s)	2.87	3.56	989.51	15.39	26.17	5.50	84.39	32.05	66.02	170.92	68.74
	Test(s)	2.82	2.91	6.7	16.15	104.83	5.97	0.03	6.54	4.54	44.08	10.84
	Parameters	89,457	281,801	4259,294	91,517	1161,637	45,225	343,614	152,857	89,357	280,968	155,188
Salinas	Train(s)	5.32	6.26	1484.56	25.05	59.15	11.28	61.57	19.64	39.81	186.02	37.04
	Test(s)	14.40	9.75	7.43	25.77	241.89	9.41	0.02	3.83	3.70	56.35	9.82
	Parameters	313,964	551,176	4269,394	155,008	2636,997	86,456	369,009	166,890	102,942	281,878	175,066
WHU-Hi-HongHu	Train(s)	5.82	6.03	-	168.65	101.18	25.61	184.19	86.91	100.95	1350.95	254.58
	Test(s)	114.40	109.83	-	171.41	2508.45	73.87	0.06	16.94	15.08	459.63	52.11
	Parameters	585,570	722,398	-	198,019	3374,977	89,342	386,703	176,244	111,912	282,658	189,430

Note: "-" means that there are no experiment.

version of HGNNConv, and its performance is also superior in our model.

C. Complexity Analysis

To analyze the complexity of HGCN-MHF, we compared the running time and parameter size of 3D-CNN, SSRN, DCFSL, S3Net, DPN-HRA, SPFormer, AMGCFN, CEGCN, WFCG, and GTFN under five labeled samples in four datasets. As seen in Table XIV, the running time, including training and testing, of our HGCN-MHF is moderate and falls within an acceptable range compared to the other networks. It is worth noting that the running time of our HGCN-MHF method increases as the graph size increases. The increase in running time with larger graph sizes is attributed to the rapid growth in the number of hyperedges. This results in higher computational costs and longer execution times for the convolution

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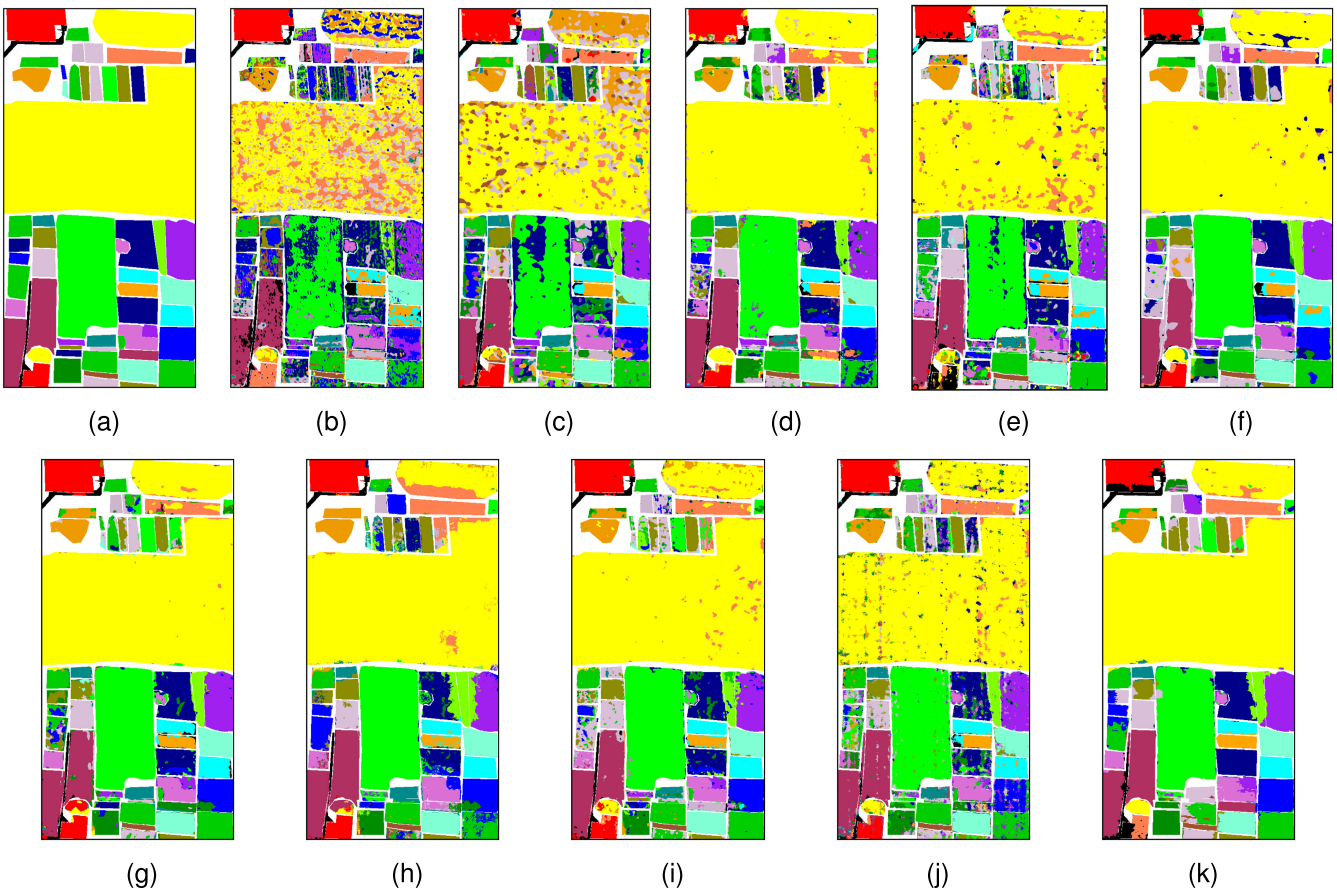


Fig. 14. Classification maps obtained by different methods for WHU-Hi-HongHu. (a) Ground truth. (b) 3D-CNN (68.83%). (c) SSRN (70.29%). (d) S3Net (81.97%). (e) DPN-HRA (75.54%). (f) SPFormer (84.64%). (g) AMGCFN (85.73%). (h) CEGCN (82.56%). (i) WFCG (84.76%). (j) GTFN (80.44%). (k) HGCN-MHF (88.86%).

operation in spatial domain HGC. Nevertheless, the parameters of HGCN-MHF remain small, making it suitable for limited training samples classification.

V. Conclusion

In this article, we propose a novel HGCN-MHF for HSI classification with limited samples. First, we introduce a CNN branch aimed at capturing pixel-level feature with different receptive fields, incorporating DST for spectral feature denoising and dimensionality reduction, and CMC for extracting feature with diverse kernel sizes in both spatial and spectral domains. Second, we propose an HGCN branch that combines MHF module and HGC to generate a comprehensive hypergraph using the spatial topological information and spectral features inherent in the segmented regions, and extract the superpixel-level feature. Finally, we design an SWFF strategy that combines the two branches, which performs adaptive weighting based on the calculated scores, and then predicts the classification graph using the softmax function.

Although we have obtained encouraging experimental results, we will continue to work on improving the proposed method. Currently, the effectiveness of the HGCN branch heavily depends on the quality of the segmentation result, which is susceptible to noise in HSI. To optimize the hypergraph structure, the incorporation of dynamic HGNNs becomes crucial, meaning that the network adaptively updates the hypergraph utilizing the learned knowledge. The optimization is embodied in two aspects, that is, suppressing noise to the segmented HSI, iteratively optimizing the segmented region, and rebuilding the hyperedge to obtain a more reasonable higher order adjacency relationship. Our future work will explore the integration of dynamic mechanisms into the proposed method, aiming to improve noise suppression and overall performance in HSI classification.

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